

First Year

FUNDAMENTAL OF NURSING



THEORITICAL BOOK

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List of abbreviation

Abbreviation	Meaning
ANA	American Nursing Association
RN	Registered nurse
ICN	International Council of Nurses
CDC	Centers for Disease Control and Prevention
WHO	World Health Organization
NANDA	North American Nursing Diagnosis Association
ECF	Extracellular fluid
I & O	Intake & Output
ABG	Arterial Blood Gases
STD	Sexual Transmitted Disease
GI	Gastro intestinal
ADL	Activity of Daily Living
RTA	Road Traffic Accident
ER	Emergency
OPD	Outpatient Department
ICU	Intensive Care Unit
AED	Automated External Defibrillator
CPR	Cardiopulmonary Resuscitation
PPE	Personal Protective Equipment

Chapter one (Foundation of Nursing Practice)

Learning objectives:

At the end of the chapter the students will be able to:

1. Describe the historic background of nursing, definitions of nursing, and the status of nursing as a profession and as a discipline.
2. Discuss the development of professional nursing roles.
3. Explain the aims of nursing as they interrelate to facilitate maximal health and quality of life for patients.
4. Identify purpose of patient safety
5. List factors influencing patient safety
6. Illustrate international patient safety goal

Introduction

Nursing is an art and a science. As a professional nurse you will learn to deliver care artfully with compassion, caring, and respect for each patient's dignity and personhood. As a science, nursing practice is based on a body of knowledge that is continually changing with new discoveries and innovations.

Definition of nursing.

Nursing is the protection, promotion, and optimization of health and abilities; prevention of illness and injury; alleviation of suffering through the diagnosis and treatment of human responses; and advocacy in health care for individuals, families, communities, and populations. **(ANA, 2018).**

Definition of a Nurse.

The nurse is a person who has completed a program of basic, generalized nursing education and is authorized by the appropriate regulatory authority to practice nursing in his/her country.

Aim of nursing:

- Four broad aims of nursing practice can be identified in the definitions of nursing:
 1. To promote health
 2. To prevent illness
 3. To restore health
 4. To facilitate coping with disability or death
- To meet these aims, the nurse uses knowledge, skills, and critical thinking to give care in a variety of traditional and expanding nursing roles.

The nursing profession and its characteristics:

A variety of career opportunities are available in nursing, including clinical practice, education, research, management, administration, and even entrepreneurship. As a student it is important for you to understand the scope of professional nursing practice and how nursing influences the lives of your patients, their families, and their communities. The patient is the center of your practice.

Types of standards for nursing

» Standards of Practice.

The nursing process is the foundation of clinical decision making and includes all significant actions taken by nurses in providing care to patients.

- **Assessment:** The registered nurse collects comprehensive data pertinent to the patient's health and/or the situation.
- **Diagnosis:** The registered nurse analyzes the assessment data to determine the diagnoses or issues.
- **Outcomes Identification:** The registered nurse identifies expected outcomes for a plan individualized to the patient or the situation.
- **Planning:** The registered nurse develops a plan that prescribes strategies and alternatives to attain expected outcomes.
- **Implementation:** The registered nurse (RN) implements the identified plan.
- **Evaluation:** The nurse evaluates progress toward attainment of outcomes.

» **Standards of Professional Performance:**

The **American Nurses Association** ANA Standards of Professional Performance describes a competent level of behavior in the professional role (ANA,2010). The standards provide a method to assure patients that they are receiving high-quality care.

1. **Ethics:** The registered nurse practices ethically.
2. **Education:** The registered nurse attains knowledge and competency that reflects current nursing practice.
3. **Evidence-Based Practice and Research:** The registered nurse integrates evidence and research findings into practice.
4. **Quality of Practice:** The registered nurse contributes to quality nursing practice.
5. **Communication:** The registered nurse communicates effectively in all areas of practice.
6. **Leadership:** The registered nurse demonstrates leadership in the professional practice setting and the profession.
7. **Collaboration:** The registered nurse collaborates with health care consumer, family, and others in the conduct of nursing practice.
8. **Professional Practice Evaluation:** The registered nurse evaluates her or his own nursing practice in relation to professional practice standards and guidelines, relevant statutes, rules, and regulations.
9. **Resources:** The registered nurse uses appropriate resources to plan and provide nursing services that are safe, effective, and financially responsible.
10. **Environmental Health:** The registered nurse practices in an environmentally safe and healthy manner.

The role of the nurse and its functions:

Nurses provide care and comfort for patients in all health care settings and concern for meeting their patient's needs. These roles are described as the following: -

- 1-Direct care provider/ caregiver:** As a caregiver, the nurse integrates the roles of communicator, teacher, counselor, leader, researcher, advocate, and collaborator to meet physical, emotional, intellectual, sociocultural, and spiritual needs for all patients.
- 2- Communicator:** The use of effective interpersonal and therapeutic communication skills to establish and maintain helping relationships with patients of all ages in a wide variety of healthcare settings.
- 3-Teacher/Educator:** The use of communication skills to assess, implement, and evaluate individualized teaching plans to meet learning needs of patients and their families.
- 4- Counselor:** The use of therapeutic interpersonal communication skills to provide information, make appropriate referrals, and facilitate the patient's problem-solving and decision-making skills.
- 5-Leader:** The assertive, self-confident practice of nursing when providing care, effecting change, and functioning with groups.
- 6- Researcher:** The participation in or conduct of research to increase knowledge in nursing and improve patient care.
- 7-Advocate:** The protection of human or legal rights and the securing of care for all patients based on the belief that patients have the right to make informed decisions about their own health and lives.

8- Collaborator: The effective use of skills in organization, communication, and advocacy to facilitate the functions of all members of the healthcare team as they provide patient care.

➤ The nurse carries out these roles in many different settings, with care increasingly provided in the home and in the community. Examples for providing patients are the following:

- Hospitals
- Ambulatory surgery centers
- Emergency helicopter services
- Clinics
- Homes
- Educational programs
- Public health offices
- Doctors' offices
- Industry
- Long-term care facilities
- Mobile healthcare units
- Schools
- Offices
- Hospice
- Mental health facilities
- State health programs
- Skilled-care facilities
- Churches
- Prisons

Ethics in nursing profession

Introduction

Ethical values are essential for any healthcare provider. Ethical values are universal rules of conduct that provide a practical basis for identifying what kinds of actions, intentions, and motives are valued.

Definition of ethics:

Ethics are moral principles that govern how the person or a group will behave or conduct themselves.

Definition of nursing ethics:

systematic rules or principles governing right conduct. Each nurse, is practiced with the responsibility to adhere to the standards of ethical practice and conduct set by the profession.

Ethical principles:

Ethical principles are the basis of all nursing practice and provide a framework to help the nurse in ethical decision making.

The primary ethical principles include:

- **Beneficence:** Acting for the good and welfare of others and including such attributes as kindness and charity.
- **Nonmaleficence:** Acting in such a way as to prevent harm to others or to inflict the minimal harm possible.
- **Autonomy:** Recognizing the individual's right to self-determination and decision-making.
- **Justice:** Acting in fairness to all individuals, treating others equally and showing all individuals the same degree of respect and concern.

- **Veracity:** Being truthful, trustworthy, and accurate in all interactions with others.
- **Fidelity:** Being loyal and faithful to individuals who place trust in the nurse.
- **Integrity:** Acting consistently with honesty and basing actions of moral standards.

International Council of Nurses Code of Ethics Preamble

The ICN Code of Ethics for Nurses has four principal elements that outline the standards of ethical conduct

Elements of the Code

1. Nurses and People

1. The nurse's primary professional responsibility is to people requiring nursing care.
2. In providing care, the nurse promotes an environment in which the human rights, values, customs and spiritual beliefs of the individual, family and community are respected.

2. Nurses and Practice

- The nurse carries personal responsibility and accountability for nursing practice, and for maintaining competence by continual learning.
- The nurse maintains a standard of personal health such that the ability to provide care is not compromised

3. Nurses and the Profession

- The nurse assumes the major role in determining and implementing acceptable standards of clinical nursing practice, management, research and education.

- The nurse is active in developing a core of research-based professional knowledge.

4. Nurses and Co-workers

- The nurse sustains a co-operative and respectful relationship with co-workers in nursing and other fields.
- The nurse takes appropriate action to safeguard individuals, families and communities when their health is endangered by a co-worker or any other person.

Patient Safety

Safety, often defined as freedom from psychological and physical injury, is a basic human need. Health care provided in a safe manner and a safe community environment is essential for a patient's survival and well-being.

Definition of patient safety:

The world health organization (**WHO**) defined patient safety as the prevention of errors and adverse effects to patients associated with health care.

Purpose of patient safety:

It aims to prevent and reduce risks, errors and harm that occur to patients during provision of health care. A cornerstone of the discipline is continuous improvement based on learning from errors and adverse events.

Factors influencing patient safety

- Age and Development
- **Lifestyle** factors that place individuals at risk for injury include unsafe environments.
- **Mobility and Health Status** Alterations in mobility related to paralysis, muscle weakness, diminished balance, and lack of coordination place clients at risk for injury.
- **Sensory Perceptual Alterations** Accurate sensory perception of environmental stimuli is vital to safety.
- **Emotional State** states can alter the ability to perceive environmental hazards.

- **Ability to Communicate** Individuals with diminished ability to receive and convey information are at risk for injury.
- **Environmental Factors** Client safety is affected by the healthcare setting.
- **Cognitive Awareness** is the ability to perceive environmental stimuli and body reactions and to respond appropriately through thought and action.

The Joint Commission National Patient Safety Goals Effective January 2023 for the Hospital Program

Goal: Improve the accuracy of patient identification.

- Use at least two patient identifiers when administering medications, blood, or blood components; when collecting blood samples and other specimens for clinical testing; and when providing treatments or procedures.

Goal: Improve the effectiveness of communication among caregivers.

- Report critical results of tests and diagnostic procedures on a timely basis.

Goal: Improve the safety of using medications.

- Label all medications, medication containers, and other solutions on and off the sterile field in perioperative and other procedural settings.
- Reduce the likelihood of patient harm associated with the use of anticoagulant therapy.
- Maintain and communicate accurate patient medication information.

Goal: Reduce patient harm associated with clinical alarm systems.

- Improve the safety of clinical alarm systems.

Goal: Reduce the risk of health care-associated infections.

- Comply with either the current Centers for Disease Control and Prevention (CDC) hand hygiene guidelines or the current World Health Organization (WHO) hand hygiene guidelines.
- Implement evidence-based practices to prevent healthcare associated infections due to multidrug-resistant organisms in acute care hospitals and in nursing care centers.
- Implement evidence-based practices to prevent central line–associated bloodstream infections (CLABSI).
- Implement evidence-based practices for preventing surgical site infections.
- Implement evidence-based practices to prevent indwelling catheter-associated urinary tract infections (CAUTI).

Goal: Reduce the risk of patient harm resulting from falls.

- Reduce the risk of falls.

Goal: Prevent health care-associated pressure ulcers (decubitus ulcers).

- Assess and periodically reassess each patient's and resident's risk for developing a pressure ulcer and take action to address any identified risks.

Goal: The organization identifies safety risks inherent in its patient population.

- Reduce the risk for suicide.
- Identify risks associated with home oxygen therapy, such as home fires.

Goal: Improve health care equity.

- Improving health care equity for the organization's patients is a quality and safety priority.

Goal: Universal Protocol for Preventing Wrong Site, Wrong Procedure, Wrong Person Surgery

- Conduct a preprocedural verification process.
- Mark the procedure site.
- A time-out is performed before the procedure.

Health and Illness Concept

Definition of Health

Defining health is difficult. The World Health Organization (WHO) defines health as a “state of complete physical, mental, and social wellbeing, not merely the absence of disease or infirmity” (WHO, 1947).

Definition of Illness

- Illness is a state in which a person’s physical, emotional, intellectual, social, developmental, or spiritual functioning is diminished or impaired.
- Cancer is a disease process, but one patient with leukemia who is responding to treatment may continue to function as usual, whereas another patient with breast cancer who is preparing for surgery may be affected in dimensions other than the physical.

Acute and Chronic Illness

- Both acute and chronic illnesses have the potential to be life threatening. An acute illness is usually reversible, has a short duration, and is often severe. The symptoms appear abruptly, are intense, and often subside after a relatively short period.
- A chronic illness persists, usually longer than 6 months, is irreversible, and affects functioning in one or more systems.

Characteristics of Acute and Chronic Illness

Description	Characteristics
Acute Illness	
Diseases that have a rapid onset and a short duration. Examples: colds, influenza, acute gastroenteritis	▪ Usually, self-limiting ▪ Responds readily to treatment ▪ Complications infrequent ▪ After illness, return to previous level of functioning
Chronic Illness	
Diseases that are prolonged, do not resolve spontaneously, and are rarely cured completely. Examples: Alzheimer's disease, Arthritis, Cancer	▪ Permanent impairments or deviations from normal. ▪ Irreversible pathologic changes ▪ Residual disability ▪ Special rehabilitation required ▪ Need for long-term medical and/or nursing management

Variables Influencing Health and Illness

1. Internal Variables

- **Developmental Stage.** A person's thought and behavior patterns change throughout life.
- **Intellectual Background.** A person's beliefs, knowledge, or incorrect information about body functions and illnesses, educational background, traditions, and past experiences.
- **Perception of Functioning.** The way people perceive their physical functioning affects health beliefs and practices.
- **Emotional Factors.** The patient's degree of stress, depression, or fear can influence health beliefs and practices.
- **Spiritual Factors.** Spirituality including the values and beliefs exercised, the relationships established with family and friends, and the ability to find hope and meaning in life.

2. External Variables

- **Family Practices.** The way that patients' families use health care services generally affects their health practices. Their perceptions of the seriousness of diseases and their history of preventive care behaviors.
- **Psychosocial and Socioeconomic Factors.** Socioeconomic and psychosocial factors increase the risk for illness and influence the way that a person defines and reacts to illness.
- **Cultural Background.** Cultural background influences beliefs, values, and customs.

Health Promotion, Wellness, and Illness Prevention

- **Health promotion** activities such as routine exercise and good nutrition help patients maintain or enhance their present levels of health.
- **Wellness** education teaches people how to care for themselves in a healthy way and includes topics such as physical awareness, stress management, and self-responsibility.
- **Illness** prevention activities such as immunization programs protect patients from actual or potential threats to health.

Levels of Preventive Care

1. Primary Prevention.

Primary prevention is true prevention; it precedes disease or dysfunction and is applied to patients considered physically and emotionally healthy. Primary prevention aimed at health promotion includes health education programs, immunizations, nutritional programs, and physical fitness activities.

2. Secondary Prevention

Secondary prevention focuses on individuals who are experiencing health problems or illnesses and are at risk for developing complications or worsening conditions. Activities are directed at diagnosis and prompt intervention, thereby reducing severity and enabling the patient to return to a normal level of health as early as possible.

3. Tertiary Prevention

Occurs when a defect or disability is permanent and irreversible. It involves minimizing the effects of a long-term disease or a disability by interventions directed at preventing complications and deterioration. Activities are directed at rehabilitation rather than diagnosis and treatment.

- **For example,** a patient with a spinal cord injury undergoes rehabilitation to learn how to use a wheelchair and perform activities of daily living independently. Care at this level helps patients achieve as high a level of functioning as possible, despite the limitations caused by illness or impairment.

Students' self-assessment

Circle the letter that corresponds to the best answer for each question .

Immunizing children against measles is an example of which of the following levels of preventive care?

- a. Primary
- b. Secondary
- c. Tertiary

Referring an HIV-positive patient to a local support group is an example of which of the following levels of preventive care?

- a. Primary
- b. Secondary
- c. Tertiary

Which of the following guidelines was developed by the American Hospital Association to enumerate the rights and responsibilities of patients while receiving hospital care?

- a. Code of Ethics
- b. Patient Bill of Rights
- c. Biomedical ethics
- d. Hospital patient advocacy

- 1. Mention Factors influencing patient safety**
- 2. Enumerate The role of the nurse and its functions.**
- 3. differentiate between Characteristics of Acute and Chronic Illness**

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Chapter Two (Communication)

Learning objective

At the end student will be able to:

- Identify the communication
- Describe the components of the communication process
- Discuss the mods (forms) of communication
- Categories the levels of communication
- Describe factors influencing the communication process
- Discuss therapeutic communication technique

➤ Introduction

Communication is a critical skill for nursing. It is the process by which humans meet their survival needs, build relationships, and experience emotions. In nursing, communication is a dynamic process used to gather assessment data, to teach and persuade, and to express caring and comfort. It is an integral part of the helping relationship.

➤ Definition of communication

Communication is the process of exchanging information and generating and transmitting meanings between two or more people. It is the foundation of society and the most primary aspect of a nurse patient interaction.

➤ Basic components of the communication process

The communication process is initiated based on a stimulus; in this case a patient need that must be addressed. The need might be due to a patient's discomfort or desire for information or to address any uncertainty the patient might be experiencing.

1. **The sender** or source (encoder) of the message is a person or group who initiates or begins the communication process.
2. **The message** is the actual communication product from the source. It might be a speech, interview, conversation, chart, gesture memorandum, or nursing note.
3. **The channel of communication** is the medium the sender has selected to send the message. The channel might target any of the receiver's senses. The message can be sent to the receiver through the following channels:
 - Auditory—spoken words and cues
 - Visual—sight, observations, and perception

- Kinesthetic—touch

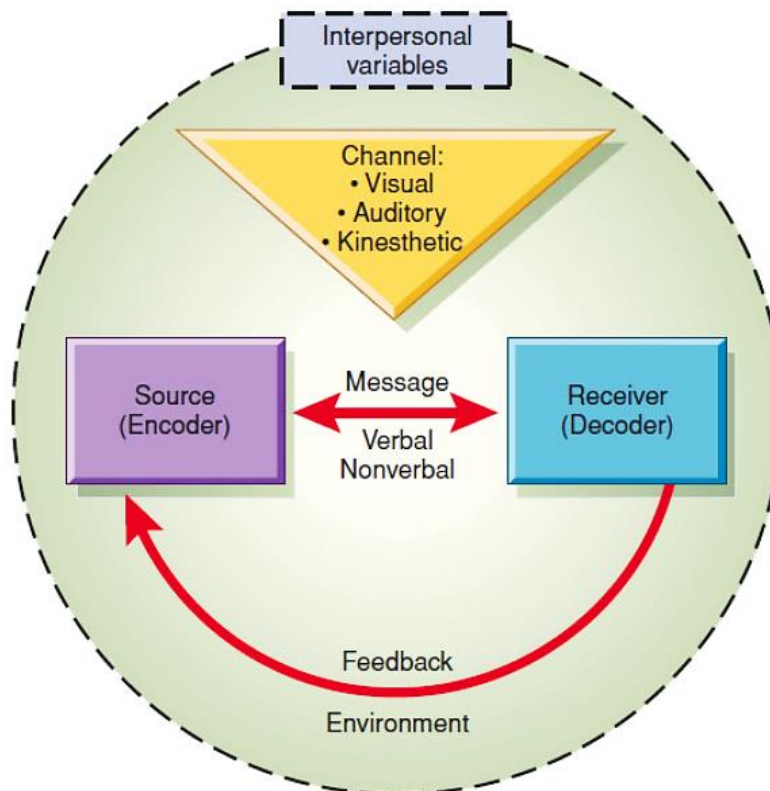
Nurses use all three of these channels to communicate with patients and other health care providers.

- 4. The receiver (decoder)** must translate and interpret the message sent and received.

To be an effective communicator, the nurse needs to be considerate of the receiver, and select a message that appeals to the patient's interests and that requires minimal effort and time to decode.

- 5. Confirmation** of the message provides **feedback** (i.e., evidence) that the receiver has understood the intended message.

Communication is a reciprocal process in which both the sender and the receiver of messages participate simultaneously.



The various components in the process of communication.

➤ The Mods (Forms) Of Communication

The sending and receiving of messages is accomplished through verbal and nonverbal communication techniques. These can occur separately or simultaneously.

1. Verbal Communication

- **Verbal communication** is an exchange of information using words including both the spoken and written word. Verbal communication depends on **language**, or a prescribed way of using words so that people can share information effectively.
- Nurses use verbal communication extensively when providing patient care, including verbal interactions with patients and family giving oral reports to other nurses and health care providers developing nursing care plans, and evaluating patient progress.

2. Nonverbal Communication

Nonverbal communication is the transmission of information without the use of words, also known as **body language**. It often helps nurses to understand subtle and hidden meanings in what the patient is saying verbally.

➤ The various forms of nonverbal communication follow.

- **Touch**

Tactile sense has been studied seriously as a form of nonverbal communication only since the 1960s. Touch is a personal behavior and means different things to different people. Familial, regional, class, and cultural influences largely shape tactile experiences.

- **Eye Contact**

Communication often begins with eye contact. A glance, for example is often an attention-getting method to open conversation. In many cultures, eye contact suggests respect and a willingness to listen and to keep communication open.

- **Facial Expressions**

The face is the most expressive part of the body. Examples of the various messages facial expressions convey are anger, joy, suspicion sadness, fear, and contempt.

- **Posture**

The way a person holds the body carries nonverbal messages. People in good health and with a positive attitude usually hold their bodies in good alignment. Depressed or tired people are more likely to slouch Posture also often provides nonverbal clues concerning pain and physical limitations.

- **General Physical Appearance**

Many illnesses cause at least some alterations in general physical appearance. Observing for changes in appearance is an important nursing responsibility for detecting illness or evaluating the effectiveness of care and therapy.

- **Sounds.**

Sounds such as sighs, moans, groans, or sobs also communicate feelings and thoughts. They have several interpretations: moaning conveys pleasure or suffering, and crying communicates happiness, sadness, or anger. Validate nonverbal messages with the patient to interpret them accurately.

3. Electronic Communication

- The Internet and a variety of social websites provide new and challenging opportunities for nurses to communicate and collaborate with other health care providers.
- The challenges of using social media include protecting patient privacy and confidentiality and preventing unintended consequences for the employer or the nurse.

- **Social Media**

Social media are web-based technologies that allow users to create share, and participate in dialogue in virtual communities and networks The availability of social media sites has dramatically changed communications among people, communities, and organizations social media networks allow nurses to share ideas, develop professional connections, access educational offerings and forums receive support, and investigate evidence-based practices.

- **E-Mail and Text Messages**

E-mail and text messages are efficient means to communicate with staff members and, in some cases, patients. The risk for violating patient privacy and confidentiality exists any time a message is sent electronically.

➤ levels of communication

Throughout our lives and the lives of our patients, communication occurs at varying levels. Nurses engage in four levels of communication during practice:

- intrapersonal communication
- interpersonal communication,
- small-group communication,
- Public communication

- **Intrapersonal communication**

is a powerful form of communication that occurs within an individual. This level of communication is also called self-talk self- verbalization or inner thought People's thoughts strongly influence perceptions, feelings, behavior, and self-concept.

- **Interpersonal communication**

is one-on-one interaction between a nurse and another person that often occurs face to face. It is the level most frequently used (in nursing situations and lies at the heart of nursing practice.)

- **Small-group communication**

is an interaction that occurs when a small number of people meet. This type of communication is usually goal-directed and requires an understanding of group dynamics.

- **Public communication**

is interaction with an audience. Nurses have opportunities to speak with groups of consumers about health-related topics, present scholarly work to colleagues at conferences, or lead classroom discussions with peers or students.

➤ **Factors influencing the communication process**

Many factors influence the communication process. Some of these are development, gender, values and perceptions, personal space, territoriality, roles and relationships, environment, congruence, interpersonal attitudes, and boundaries.

- **Development**

Language, psychosocial, and intellectual development move through stages across the lifespan. Knowledge of patient's developmental stage will allow the nurse to modify the message accordingly.

- **Gender**

From an early age, females and males communicate differently. Girls tend to use language to seek confirmation, minimize differences, and establish intimacy. Boys use language to establish independence and negotiate status within a group.

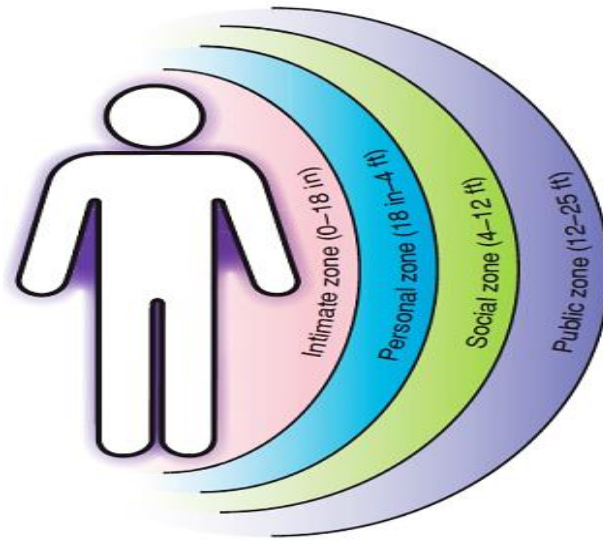
- **Values and Perceptions**

Values are the standards that influence behavior, and perceptions are the personal view of an event. Because each individual has unique personality traits, values, and life experiences, each will perceive and interpret messages and experiences differently.

- **Personal Space**

Personal space is the distance people prefer in interactions with others

1. Intimate: 0 to 12 feet
2. Personal: 12 to 4 feet
3. Social: 4 to 12 feet
4. Public: 12 feet and beyond



- **Environment**

People usually communicate most effectively in a comfortable environment. Temperature extremes, excessive noise, and a poorly ventilated environment can all interfere with communication.

- **Interpersonal Attitudes**

Attitudes convey beliefs, thoughts, and feelings about people and events. Attitudes are communicated convincingly and rapidly to others.

➤ **Therapeutic communication technique**

Therapeutic Communication Techniques are specific responses that encourage the expression of feelings and ideas and convey acceptance and respect.

1. Active Listening. Active listening means being attentive to what a patient is saying both verbally and nonverbally. Active listening facilitates patient communication.

- 2. Sharing Observations.** Nurses make observations by commenting on how the other person looks, sounds, or acts.
- 3. Sharing Empathy.** Empathy is the ability to understand and accept another person's reality, accurately perceive feelings, and communicate this understanding to the other.
- 4. Sharing Hope.** Nurses recognize that hope is essential for healing and learn to communicate a "sense of possibility" to others.
- 5. Sharing Humor.** Humor is an important but often underused resource in nursing interactions.
- 6. Sharing Feelings.** Emotions are subjective feelings that result from one's thoughts and perceptions. If individuals do not express feelings, stress and illness may worsen.
- 7. Using Touch.** Touch is one of the most potent forms of communication. Historically physical touch played a central role in healing.
- 8. Using Silence.** Silence prompts some people to talk. It allows a patient to think and gain insight.
- 9. Providing Information,** providing relevant information tells other people what they need or want to know so they are able to make decisions, experience less anxiety, and feel safe and secure.
- 10. Clarifying.** To check whether understanding is accurate, restate an unclear message to clarify the sender's meaning.
- 11. Focusing.** Focusing centers on key elements or concepts of a message. If the conversation is vague or patients begin to repeat themselves, focusing is a useful technique.

12.Paraphrasing. Paraphrasing is restating another's message more briefly using one's own words.

13.Asking Relevant Questions. Nurses ask relevant questions to seek information needed for decision-making. Ask only one question at a time and fully explore one topic before moving to another area.

14.Summarizing. Summarizing is a concise review of key aspects of an interaction. It brings a sense of satisfaction and closure to an individual conversation and is especially helpful during the termination phase of a nurse-patient relationship.

Students' self-assessment

An old woman was brought for evaluation due to increasing forgetfulness and limitations in daily function. She says to the nurse who offers her breakfast, “Oh no, I will wait for my husband. We will eat together” The therapeutic response by the nurse is:

- a. “Your husband is dead. Let me serve you your breakfast.”
- b. “I’ve told you several times that he is dead. It’s time to eat.”
- c. “You’re going to have to wait a long time.”
- d. “What made you say that your husband is alive?”

Which of the following are considered nonverbal cues? Select all that apply.

- a. Tone and rate of voice
- b. Eye contact and physical appearance
- c. Soft voice
- d. Use of touch

A subtle therapeutic technique that communicates to the patient that you are interested and want to hear more. It indicates your acceptance of the patient as a person. It usually involves nonverbal cues such as eye contact and nodding.

- a. Therapeutic relationship
- b. Moral support
- c. Minimal encouragement
- d. Act of kindness

When talking to a patient from another culture and with another language, the most effective way to communicate is:

- a. Using picture.
- b. Using gestures
- c. Using a translator
- d. Using an interpreter

Chapter Three **(Critical thinking in nursing practice)**

Learning objectives:

At the end of this chapter the student will be able to:

1. Critical thinking

- Define critical thinking
- Discuss critical thinking skills in nursing

2. Nursing process

- Define the Nursing Process.
- Identify the characteristics of Nursing Process
- List the purpose of Nursing Process
- Explain Phases of Nursing Process
- Develop nursing care process.

3. Health assessment

- Define health assessment.
- List types of assessment.
- Discuss phases of health assessment.

4. Physical examination

- Define physical examination.
- Mention purpose of physical examination.
- Describe the types used in physical examination.
- Discuss the components of physical examination.

Introduction:

Critical thinking is an essential skill for nursing students to have, because it helps them make decisions based on the available information and their past experiences and knowledge of the field. It also allows nurses to plan before making any intervention to be most effective as possible. The nurse uses critical thinking and problem-solving skills to determine the underlying cause of the patient's symptoms. Critical thinking abilities compel nurses to challenge assumptions, question the context, look for new ways of doing and thinking, and consider, filter, and evaluate ideas or solutions.

Critical Thinking Skills in Nursing

Definition:

Critical thinking is an active form of thinking that is used within the healthcare setting by nurses. It involves taking data from situations and analyzing the information to help formulate the best course of action objectively.

Critical thinking skills:

Critical thinking skills that will be utilized in nursing are based on the following cognitive skills:

1. **Recognition** - when a nurse understands that there is a problem
2. **Questioning** - when a nurse decides whether safety is a concern
3. **Information gathering** - when a nurse gathers data through observations
4. **Evaluation** - when the nurse evaluates the decisions to understand which should be applied

5. **Communication** - when the decision is communicated to other healthcare professionals if needed
6. **Observation** - a nurse's assessment that forms the basis for all decisions
7. **Analysis** - when the nurse examines the observations
8. **Interpretation** - when the nurse takes raw data and analyzes the results
9. **Reflection** - when the nurse looks back on the decisions and data to understand if the choice was sound
10. **Inference** - when the nurse draws a conclusion based on the information gathered
11. **Problem-solving** - when the nurse applies critical thinking skills to the issue at hand
12. **Decision-making** - when the nurse makes a concrete choice based on best practice

Uses of critical thinking in nursing:

critical thinking is applied by nurses through the nursing process for solving problems of patients and decision-making process with creativity to enhance the effect of the care provide to the patient.

Nursing process

Definition:

A **process** is a series of steps or acts that lead to accomplishment of some goal or purpose.

Nursing process: American Nurses Association defined as an **organized systematic** method of problem-solving steps used to identify and to manage the health problems of patients.

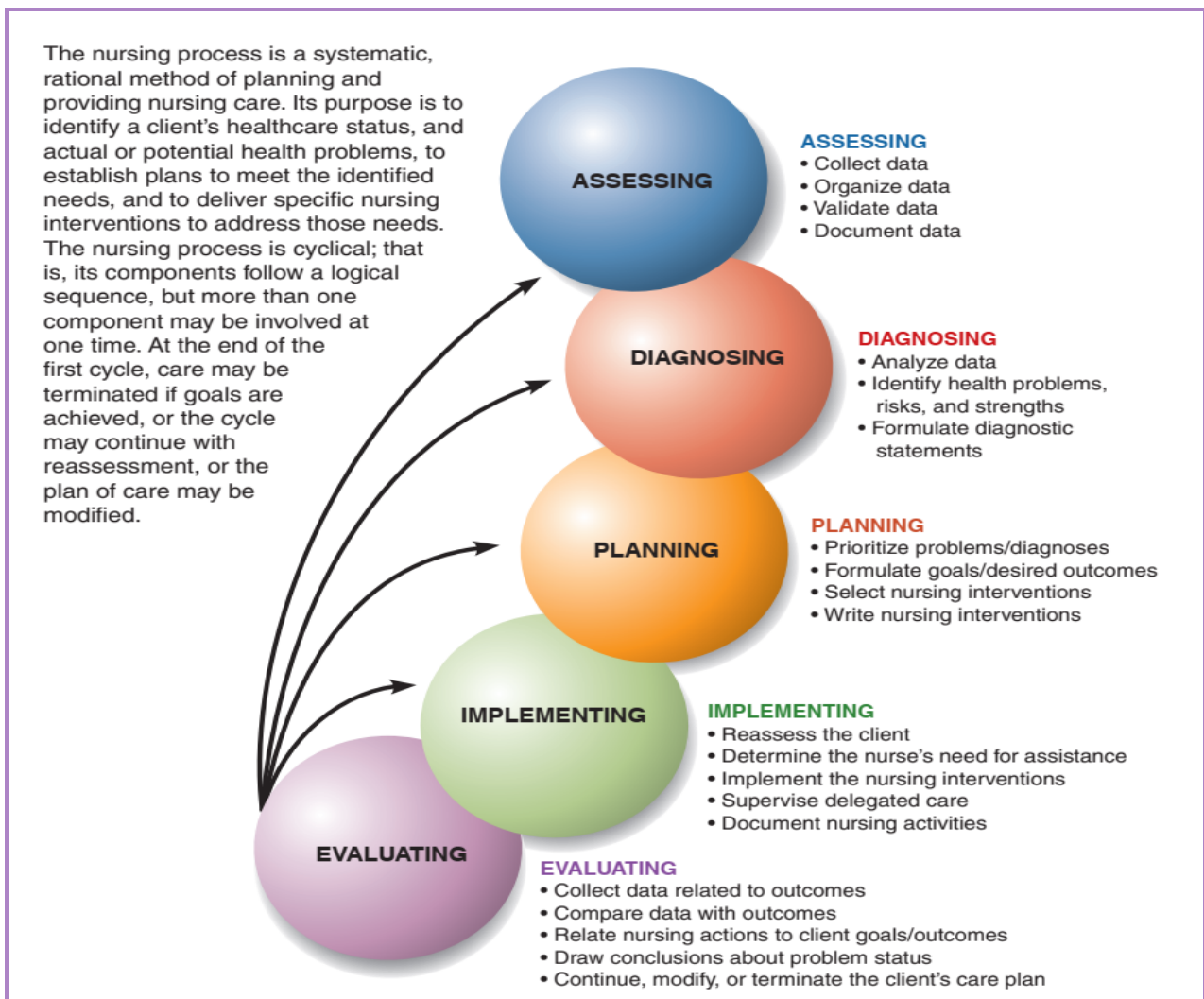


Figure (1) The Nursing Process in action

Purpose of nursing process:

- To identify a client's health status and actual or potential health care problems or needs.
- To establish plans to meet the identified needs.
- To deliver specific nursing interventions to meet those needs.

Characteristics of Nursing Process

- Continuous & Dynamic nature
- patient centered
- Focus on problem solving and decision making
- Interpersonal and collaborative style
- Universal applicability
- Use of critical thinking and clinical reasoning.

Components of nursing process

It involves:

- Assessment (data collection).
- Nursing diagnosis.
- Planning.
- Implementation.
- Evaluation.

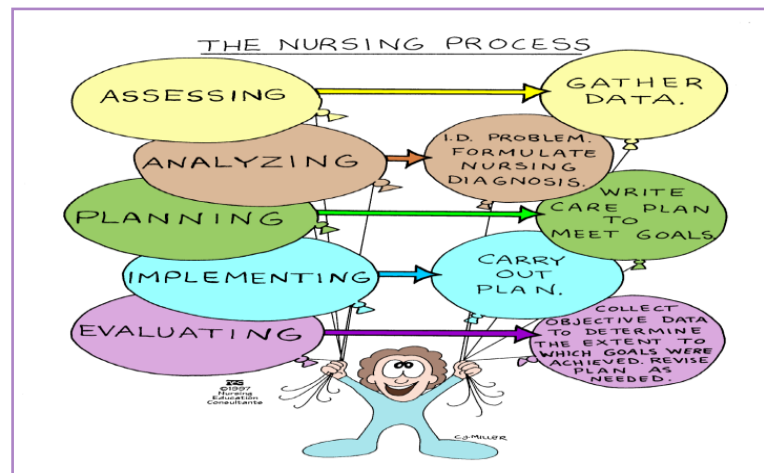


Figure (2) Components of the Nursing Process

1. ASSESSMENT

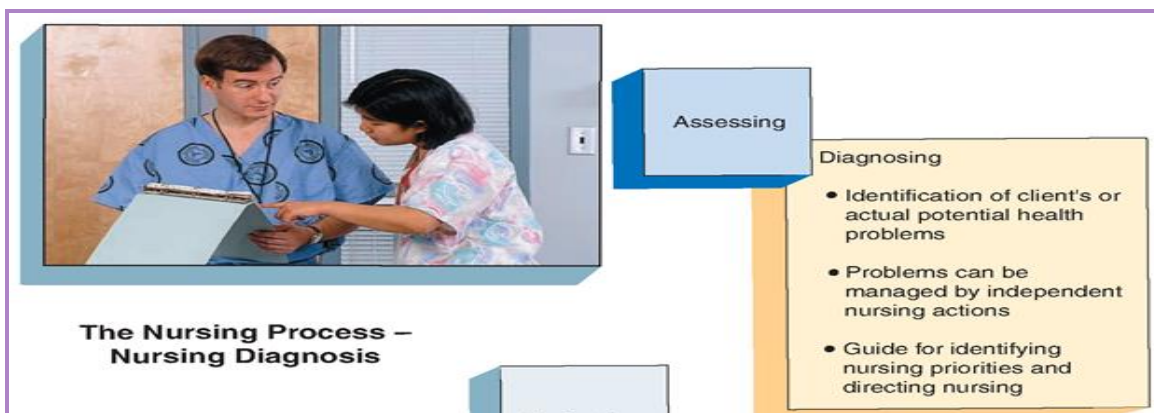
Definition: Assessment is the systematic and continuous collection, organization, validation, and documentation of data.

During this step, the nurse gathers information about the patient's health status, including physical, psychological, and social factors using different assessment techniques such as head-to-toe physical assessment. This data is used to form an accurate diagnosis and develop a care plan.



2. Diagnosis

Diagnosis is the second phase of the nursing process. In this phase, nurses use critical thinking skills to interpret assessment data to identify client problems.



A. North American Nursing Diagnosis Association (NANDA) define or refine **nursing diagnosis** is: “a clinical judgment concerning a human response to health conditions/life processes, or a vulnerability for that response, by an individual, family, group, or community.”

Purposes of the Nursing Diagnosis

- B. Identifies nursing priorities
- C. Directs nursing interventions to meet the client's high-priority needs
- D. Provides a common language and forms a basis for communication and understanding between nursing professionals and the healthcare team.

Difference between medical and nursing diagnosis

- **A nursing diagnosis:** Is a clinical judgment about individual, family, or community responses to actual or potential health problems.
- **A medical diagnosis:** Is a clinical judgment by the physician that identifies or determines a specific disease, condition, or pathological state.

Medical Diagnosis	Nursing Diagnosis
Focuses on illness, injury, or disease process.	Focuses on client's responses to actual or potential health problems or life processes.
Remains constant until a cure is affected or client dies.	Changes as the client's response and/or the health problem changes.
Example:(Lung cancer, heart failure, Brain tumor, Bronchial asthma).	Example: (Nausea, Acute pain, Anxiety, Impaired physical mobility, Ineffective breathing pattern, Risk for imbalanced fluid volume).

Types of nursing diagnosis:

1. **An Actual Diagnosis** is a client problem that is present at the time of the nursing assessment.
 - **Examples** are ineffective breathing pattern and anxiety.
2. **Risk (Potential)Nursing Diagnosis** is a clinical judgement that a problem does not exist, but the presence of risk factors indicates that a problem may develop if adequate care is not given.

- **For example**, all people admitted to hospital have some possibility of acquiring an infection; however, a client with diabetes or a compromised immune system is at higher risk than others.

3. Wellness Nursing Diagnosis

“Describes human responses to levels of wellness in an individual, family or community that have a readiness for enhancement”.

- **Examples** of wellness diagnoses would be readiness for enhanced spiritual well-being or readiness for enhance family coping.

4. Possible Nursing Diagnosis

- Is one in which evidence about a health problem is incomplete or unclear.
- Requires more data either to support or to refute it.
- **For example**, an elderly widow who lives alone is admitted to the hospital. The nurse notices that she has no visitors and is pleased with attention and conversation from the nursing staff.

5. Syndrome Nursing Diagnosis

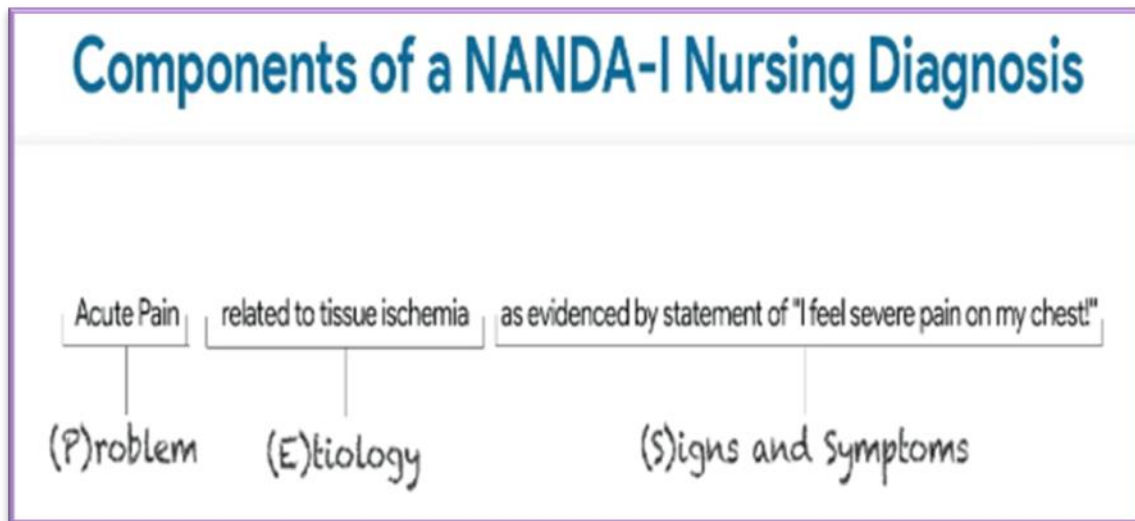
Is a diagnosis in which is associated with a cluster of other diagnoses.

For example, Clusters of diagnoses associated with this syndrome include impaired physical mobility, risk for impaired tissue integrity, risk for activity intolerance, risk for constipation, risk for infection, risk for injury, risk for powerless, impaired gas exchanged, and so on.

Components of a NANDA Nursing Diagnosis has three components:

- 1- **Problem (P)**: statement of the client's health problem (NANDA label)
- 2- **The etiology** component of a nursing diagnosis identifies causes of the health problem.
- 3- **The defining characteristics** (are the cluster of signs and symptoms that indicate the presence of health problem.

Example: Acute pain related to tissue ischemia as evidenced by patient discomfort sever pain on my chest and pain scale.



3. Planning

- Planning involves decision making and problem solving.
- Establishing client goals/desired outcomes
- Setting priorities
- Selecting nursing interventions and activities
- Writing individualized nursing interventions on care plans.

4. Implementation

The process of implementation includes;

- Implementing the nursing interventions
- Documenting nursing activities

5. Evaluation

The evaluation includes;

- Comparing the data with desired outcomes
- Continuing, modifying, or terminating the nursing care plan.

Example:

Assessment	Nursing diagnosis	Goal / expected outcome	Intervention	Rational	Evaluation
Fever body temperature is 38.5c	Body temperature alteration (fever) related to infection as manifested by elevated body temp more than normal value(36.5-37.4)c	Decrease body temperature to normal value (36.5-37.5c)	1- measure body temperature every 1 hours 2- make compresses 3- decrease body movements & activities 4- increase fluid intake 5- administration antipyretic medication as prescribed	1- to determine degree of fever 2- to decrease body temperature 3- to avoid increase of body temperature 4- to prevent dehydration 5- to relieve fever	1-Determine Patient normal body temperature within normal level. 2- goal met.

Health Assessment& physical examination

Introduction

Assessment is a key component of nursing practice, required for planning and provision of patient and family centered care. Assessment can be called the “base or foundation” of the nursing process. Nursing assessments are critical to the job of being a nurse, and there are several different types of assessments that nurses need to be able to perform.

Assessment definition

is a process where a nurse gathers, sorts and analyzes a patient's health information using evidence informed tools to learn more about a patient's overall health, symptoms and concerns



Purpose of assessments:

- Obtain baseline data and expand the data base from which subsequent phases of the nursing process can evolve
- To identify and manage a variety of patient problems (actual and potential)
- Evaluate the effectiveness of nursing care
- Enhance the nurse-patient relationship
- Make clinical judgments

Types of assessment:

1. Initial comprehensive assessment:

- Provide baseline of client data including a complete health history and current needs assessment. general appearance, physical examination and vital signs.
- Usually completed upon admission to health care agency.

➤ Problem Focused or Ongoing assessment:

- Is limited to potential health care risks, or a particular need.
- Follow up, or monitoring of specific problems.
- Expands the database and allow the nurse to confirm the validity of data obtained during the initial assessment.
- Systematic monitoring allows the nurse to determine the clients' response to nursing interventions and to identify any other problems.

➤ Emergency assessment:

Primary assessment includes (Airway, Breathing, Circulation and Disability) for determining life threatening problems.

➤ Time-lapsed Reassessment

Time lapsed reassessment, takes place after several months to the initial assessment to evaluate any changes in the clients' health status (e.g., periodic outpatient clinic visits, home health visits, health and development screenings) to compare the client 's current status to baseline data previously obtained.

Steps OR phases of assessment:

- A. Data collection from a variety of sources.
- B. Data validation.

- C. Organizing the data.
- D. Data interpretation (Data analysis).
- E. Making initial inferences or impressions.
- F. Recording or reporting data.

1. Sources of Data collection:

- **Primary sources:** the client should be considered the primary source of data. As much information as possible should be gathered from the client, using both interview techniques and physical examination skills.
- **Secondary sources:** data source from other than the client is considered secondary sources (family members, other health care providers, and medical records).

➤ Methods of data collection:

Data are obtained through:

- Interviews- patient, nurses, support persons
- Physical examinations
- Observations
- Review of records and diagnostic reports
- Collaboration with colleagues

➤ Types of Data collection:

- **Subjective data** (also called symptoms): are data from the client's point of view (provided verbally by the patient) and include feelings, perceptions, and concerns. Interview is the primarily method of collecting subjective information.

Examples of subjective information:

- A. I have had pains in my legs for three days ago.
 - B. I have had headache, nausea, vomiting, and dizziness for three hours ago.
 - C. I have had anxiety from surgery.
- **Objective data** (also called signs): are observable and measurable data that are obtained through both physical examination and the result of laboratory and diagnostic tests.

Examples of objective information include:

- Temperature (37.3°C), Pulse rate (100 b/m), Respiration (18 T/m), Blood pressure (130/76 mm/hg).
- Positive bowel sounds.
- Flushed face.

2- Validating the data:

Objective information may add to or validate subjective information. Validation is a critical step in data collection to avoid omissions, prevent misunderstandings, and avoid incorrect inferences and conclusions.

3- Organizing the data:

Data that are collected must be organized to be useful to the health care professional collecting the data as well as others involved with the client's care.

4- Interpreting the data: when data are placed in clusters the nurse can:

1. Distinguish between relevant and irrelevant data.
2. Determine whether and where there are gaps in the data.
3. Identify patterns of cause and effect.

5- Documenting the data:

Assessment data must be **recorded** and **reported**. The nurse must make a judgment about which data are to be reported immediately and which data need only to be recorded at that time.

Definition of documentation:

Any written or electronically generated information about a client that describes the care or service provided to that client as the administration of tests, procedures, treatments, and client education.

Definition of report:

A report is oral, written, or computer-based communication intended to convey information to others. For instance, nurses always report on clients at the end of a hospital work shift.

Definition of record:

A record, also called a chart or client record, is a formal, legal document that provides evidence of a client's care and can be written or computer based.

Purpose of health care documentation

1. Professional Responsibility and Accountability
2. To facilitate communication
3. To promote good nursing care
4. To meet professional and legal standards
5. Education
6. Research
7. Auditing and Monitoring

Principles of documentation

1. Date & Time

- Document date and time of each recording.
- Record time in conventional manner (E.g. 9am, 6pm etc)
- Avoid recording in advance.

2. Legibility & Signature

- Entries must be legible and easy to read.
- Writing must be clear and sign.

3. Correcting spelling & Terminology

- Very important in recording numbers and medical terms

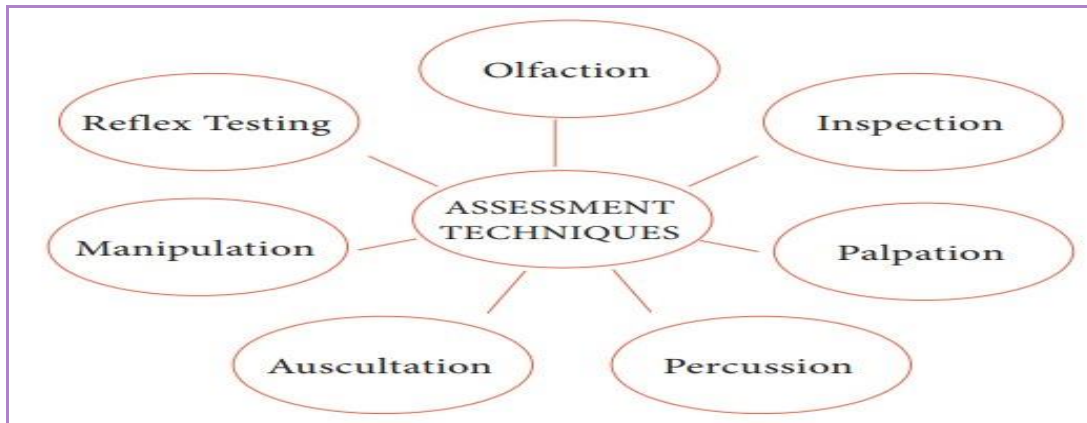
4. Completeness & appropriates

Examples of records

Form	Information
Admission (face) sheet	• Legal name, birth date, age, gender • Social Security number • Address / Marital status • Date, time, and admitting • Diagnosis • Food or drug allergies • Name of admitting primary care provider • Insurance information.
Initial nursing assessment	• Findings from the initial nursing history and physical health assessment
Graphic record	• Body temperature, pulse rate, respiratory rate, blood pressure, daily weight, and special measurements such as fluid intake and output and oxygen saturation
Daily care record	• Activity, diet, bathing, and elimination records

Special flow sheets	• Examples: fluid balance record, skin assessment
Medication record	• Name, dosage, route, time, date of regularly administered medications. • Name or initials of individual administering the medication
Nurse's notes	• Pertinent assessment of client • Specific nursing care including teaching and client's responses • Client's complaints and how client is coping
Medical history and physical examination	• Past and family medical history, present medical problems, differential or current diagnoses, findings of physical examination by the primary care provider
Physician's order form	• Medical orders for medications & treatments.
Physician's progress notes	• Medical observations, treatments & client progress.
Consultation records	• Reports by medical and clinical specialists
Diagnostic reports	• Examples: laboratory reports, x-ray reports, CT scan reports
Consultation reports	• Physical therapy, respiratory therapy
Client discharge plan and referral summary	• Started on admission and completed on discharge; includes nursing problems, general information, and referral data

Assessment Techniques:



1- **Inspection** – critical observation (always first)

1. Take time to “observe” with eyes, ears, nose.
2. Use good lighting
3. Look at color, shape, symmetry, position
4. Observe for odors from skin, breath, wound
5. Develop and use nursing instincts
6. Inspection is done alone and in combination with other assessment techniques.

2- **Palpation** – light and deep touch

- a. Back of hand (dorsal aspect) to assess skin temperature
- b. Fingers to assess texture, moisture, areas of tenderness
- c. Assess size, shape, and consistency of lesions and organs.

3- **Percussion** – sounds produced by striking body surface

- Produces different notes depending on underlying mass (dull, resonant, flat, tympanic)
- Used to determine size and shape of underlying structures by establishing their borders and indicates if tissue is air-filled, fluid-filled, or solid
- Action is performed in the wrist.

- 4- **Auscultation** – listening to sounds produced by the body
 - a. Direct auscultation – sounds are audible without stethoscope
 - b. Indirect auscultation – uses stethoscope
- 5- **Manipulation:** It means moving with the body parts. It reveals rigidity, difficulty (or) discomfort in moving the body parts.
- 6- **Reflex testing:** Means automatic response to a given stimulus. It reveals reflex is present, or not present, strength and movements of hands and legs.
- 7- **Olfaction:** It means sense of smell (Odor). It reveals the nature of disease condition of the patient.

NOTE: Assessment does not end with the initial interview and physical examination. Assessment is dynamic and continues with each nurse-client interaction.

Physical Examination

Definition

is the process of evaluating objective anatomic findings through the use of observation, palpation, percussion, and auscultation.

Purpose:

1. check for possible diseases so they can be treated early
2. identify any issues that may become medical concerns in the future
3. update necessary immunizations
4. ensure that you are maintaining a healthy diet and exercise routine

Types of physical examination

- **Comprehensive initial assessment** (e.g., when a client is admitted to a healthcare agency).
- **Focused examination of a body system** (e.g., the cardiovascular system) or body area (e.g., the lungs, when difficulty with breathing is observed).
- **Functional assessment** that examines one or more aspects of the client's abilities (e.g., nutrition and metabolism, elimination, or sleep and rest).

Component of The Physical Examination include:

- General appearance exam
- Vital signs
- Skin, hair and nail exam
- Head and Neck exam
- Neurologic exam
- Chest and Cardiovascular exam
- Abdominal exam
- Musculoskeletal exam
- Back and mobility
- Attachment and documentation

Examples:

Situation	Physical Assessment
Client complains of abdominal pain.	Inspect, auscultate, percuss, and palpate the abdomen; assess vital signs.
Client is admitted with a head injury.	Assess level of consciousness using Glasgow Coma Scale, assess pupils for reaction to light and accommodation; assess vital signs.
The client has just had a cast applied to the lower leg.	Assess peripheral perfusion of toes, capillary blanch test, pedal pulse if able, and vital signs.
The client's fluid intake is minimal.	Assess tissue turgor, fluid intake and output, and vital signs.



Student' Self-Assessment

Choose the correct answer:

This step of the nursing process includes the systematic collection of all subjective and objective data about the client in which the nurse focuses on the client- physical, psychological, emotional, sociocultural, and spiritual. Name this step

1. Assessment
2. Planning
- 3- evaluation
4. Diagnosis

Once a nurse assesses a client's condition and identifies appropriate nursing diagnoses, a:

1. List of priorities is determined.
2. Review of the assessment is conducted with other team members.
3. Plan is developed for nursing care.
4. Physical assessment begins.

Mr. Jones comes into the physicians' office with his wife. During the initial interview with the nurse assessing the reason for the visit, the wife says that her husband has been feeling bad for three days. The nurse knows this is what type of data?

1. Secondary subjective data
2. Primary subjective data
3. Secondary objective data
4. Primary objective data

Mrs. Johnson is a 62-year-old female who was admitted to the hospital with a complaint of severe headache and dizziness. Her skin is warm and dry. She has a past medical history of hypertension for the past 5 years, and she has been on antihypertensive medications. She reports occasional non-adherence to her medications due to forgetfulness. Upon admission, Mrs. Johnson's vital signs are as follows: blood pressure (BP) 180/100 mmHg, heart rate (HR) 88 beats per minute (BPM), respiratory rate (RR) 18 breaths per minute, and temperature 38.6°C.

1. What are the subjective and objective data of Mrs. Johnson?
2. Develop a plan of care based on patient needs and priorities
3. What patient education should the nurse provide to Mrs. Johnson regarding hypertension management?

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Chapter Four

A physiological Basis for nursing practice

Learning objectives:

At the end of this chapter student will be able to:

- Define of the need& human need
- Discuss Maslow's hierarchy theory
- Explain physiological needs

Introduction

Physiological or Basic needs are common to all people; thus, basic needs are universal. Individuals of all cultures have basic needs; in other words, basic needs are transcultural— across all cultures. Needs can be satisfied or they can be blocked during times of illness. Nursing is concerned with helping clients meet their physiological, spiritual, and psychological needs. Illness or risk for illness occurs when people are unable to satisfy one or more of their basic needs independently. Nursing care will center on assisting people to meet these needs. Nursing also involves helping people to avoid risks or threats to their basic human needs.

Terminology:

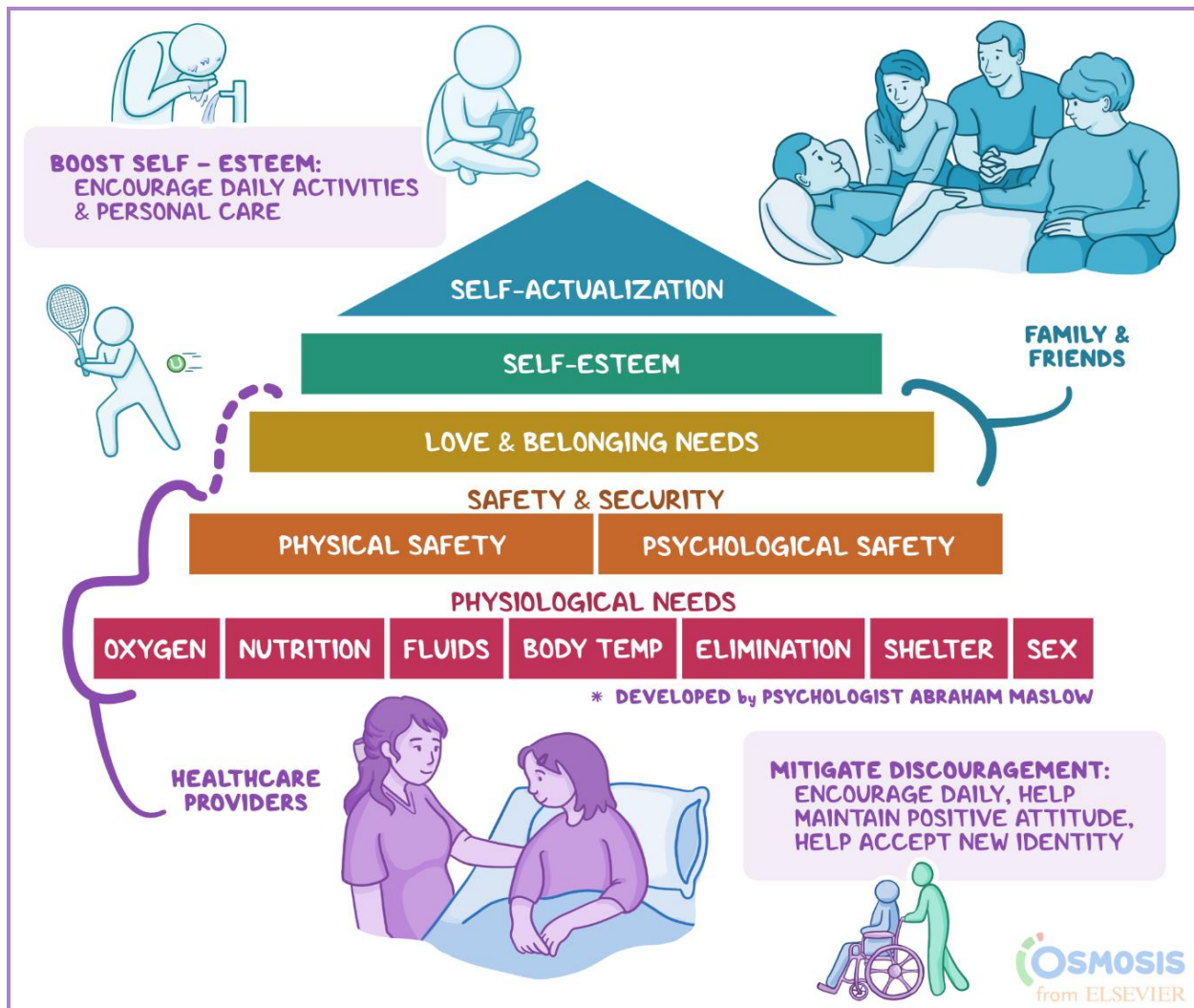
Need: A need is something that is desirable, useful or necessary.

Human needs: are physiologic and psychological conditions that an individual must meet to achieve a state of health or well-being.

Common theory related to basic needs: Maslow's hierarchy theory of Basic needs for Human Being

Maslow's hierarchy theory:

Maslow's hierarchy of needs is a five-tier list of human needs developed by Abraham Maslow, an American psychologist, often presented as a pyramid with the fundamental needs at the bottom and the higher-level needs at the top.



- Maslow defined the basic needs of all people as a progression from simple physical needs, or survival needs, to more complex ones, called aesthetic needs.

Progression from physical needs to aesthetic needs:

1. Physiologic (physical) needs

1. Oxygen
2. Fluids
3. Nutrition
4. Activity

5. Elimination
6. Rest and sleep
7. Sex concept

2. Safety and Security

1. Physical safety
2. Psychological safety
3. The need for shelter and freedom from harm and danger

3. Love and belonging

1. The need to love and be loved
2. The need to care and to be cared for.
3. The need for affection: to associate or to belong
4. The need to establish fruitful and meaningful relationships with people, institution, or organization

4. Self-Esteem Needs

1. Self-worth
2. Self-identity
3. Self-respect
4. Body image

5. Self-Actualization Needs

1. The need to learn, create and understand or comprehend
2. The need for harmonious relationships
3. The need for beauty or aesthetics
4. The need for spiritual fulfillment

Physiological needs

Learning objectives:

At the end the students will be able to:

- Define physiological needs
 - List of physiological needs
 - Discuss fluid and electrolytes and acid base.
 - Explain oxygenation in human body.
 - Discuss nutrition as a basic physiological need.
 - Discuss skin integrity to human body.
 - Explain elimination in human body.
 - Discuss rest & sleep needs for human body.
 - Analyze activity as essential need for human body.
-

Introduction

Physiological needs - these are biological requirements for human survival, e.g., air, food, drink, shelter, clothing, warmth, sex, sleep. If these needs are not satisfied the human body cannot function optimally.

Definition

Physiological needs are the most basic things that everyone needs in order to survive such as food, water, sleep, medical care, and air. When these are not satisfied, human may feel sickness, irritation, pain, discomfort, etc.

Physiologic (physical) needs

1. Fluids, electrolyte and acid base	5. Rest and sleep
2. Oxygen	6. Sex concept
3. Nutrition	7. Activity
4. Elimination	

Fluid, Electrolyte and Acid base

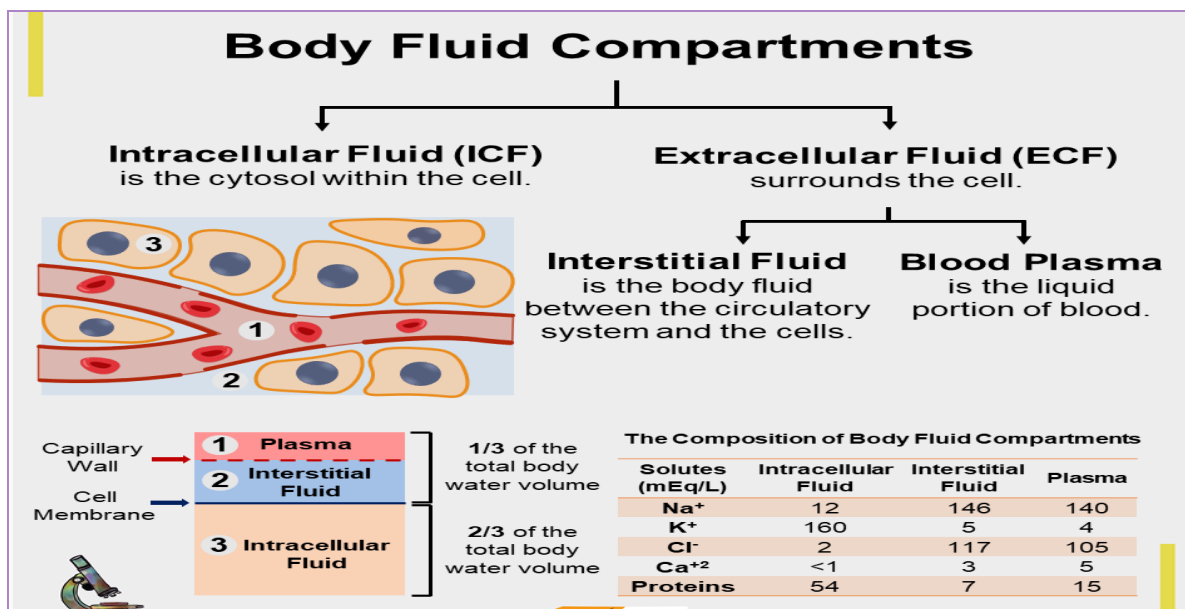
Body Fluids

Total amount of fluid in the human body which is approximately 50%-60% of the total body weight.

Body fluids compartments.

Body fluids are divided into two main compartments:

- **Intracellular fluid:** Intracellular fluid functions as a stabilizing agent for the parts of the cell, helps maintain cell shape and assists with transport of nutrients across the cell membrane, in and out of the cell.
- **Extracellular fluid.** Extracellular fluid mostly appears as interstitial tissue fluid and intravascular fluid.



Composition of body fluids:

- Water
- Dissolved solutes:
 - Organic substances:
Glucose, Amino acids, Fatty acids, Hormones & Enzymes.

- **Inorganic substances(electrolytes):**

Sodium, Potassium, Calcium, Magnesium, Chloride & Phosphate.

Normal Intake and Output

- **Daily intake.** An adult human at rest takes appropriately **2,500 ml** of fluid daily.
- **Levels of intake.** Approximate levels of intake include fluids 1, 200 ml, foods 1, 000 ml, and metabolic products 30 ml.

Daily output. Daily output should be approximately equal in intake.

- **Normal output.** Normal output occurs as urine, breathing, perspiration, feces, and in minimal amounts of vaginal secretions.

Electrolytes

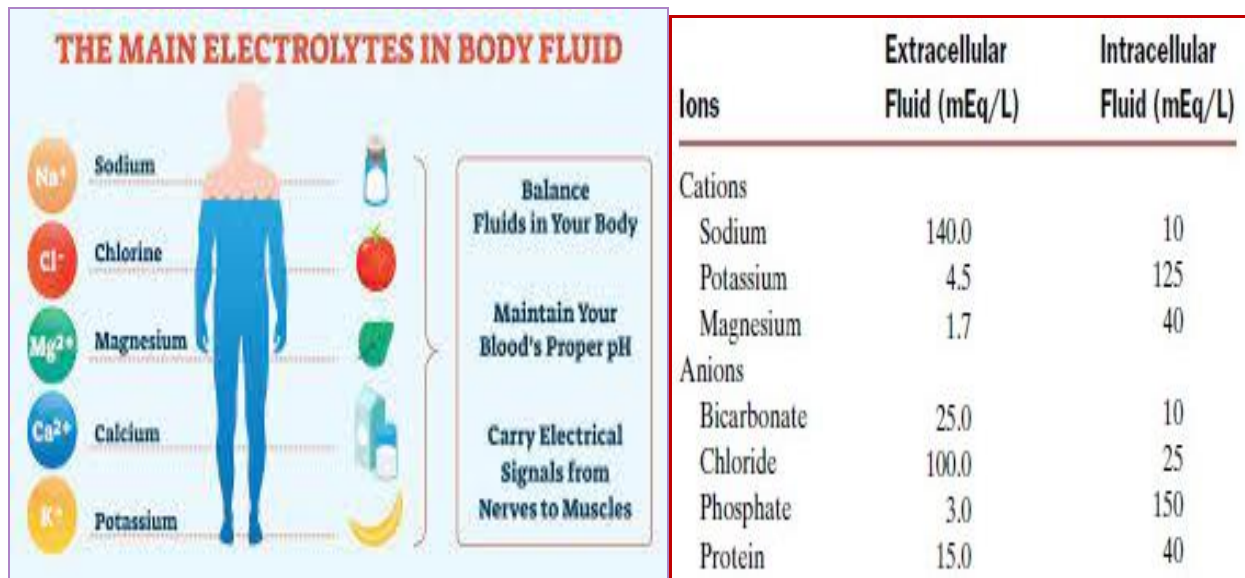
Definition:

It is a substance that will disassociate into ions when dissolved in water.

- **Origins.** Electrolytes are found in the form of inorganic salts, acids, and bases.
- **Active chemicals.** Electrolyte concentrations are measured according to their chemical activity and expressed as milliequivalents.
- **Ions.** Each chemical element has an electrical charge, either positive or negative.
- **Intracellular electrolytes.** Important intracellular electrolytes are potassium, magnesium, sulfate, and phosphate, and the most dominant cation is potassium while the most dominant anion is phosphate.

- **Extracellular electrolytes.** Important extracellular electrolytes include sodium, chlorine, calcium, and bicarbonate, and the most essential cation is sodium while chlorine is the most important anion.

Electrolytes in body fluids:



Fluid and Electrolyte Transport

Total electrolyte concentration affects the body's fluid balance.

- **The body cells.** Nutrients and oxygen should enter body cells while waste products should exit the body.
- **The cell membrane.** The cell membrane separates the intracellular environment from the extracellular environment.
- **Permeability.** The ability of a membrane to allow molecules to pass through is known as permeability.

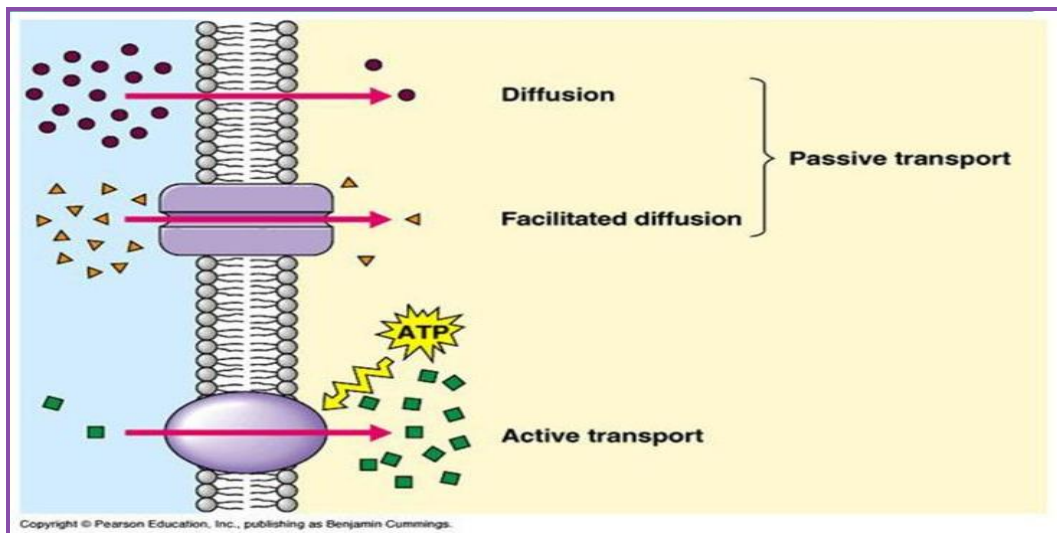
Passive Transport

Passive transport mechanisms include diffusion, osmosis, and filtration.

- **Diffusion.** Diffusion, or the process of “being widely spread”, is the random movement of molecules from an area of higher concentration to an area of lower concentration.
- **Osmosis.** Osmosis is the diffusion of a pure solvent, such as water, across a semipermeable membrane in response to a concentration gradient in situations where the molecules of a higher concentration are non-diffusible.
- **Filtration.** Filtration is the transport of water and dissolved materials concentration already exists in the cell.

Active Transport

- **Mechanisms.** Active transport mechanisms require specific enzymes and energy expenditure in the form of **adenosine triphosphate (ATP)**.



Different fluid volume disturbances that may affect an individual.

- **Fluid volume deficit** or hypovolemia occurs when the loss of ECF volume exceeds the intake of fluid.

- **Fluid volume excess** or hypervolemia refers to an isotonic volume expansion of the ECF caused by the abnormal retention of water and sodium in approximately the same proportions in which they normally exist in the ECF.

Disturbances in electrolyte balances are common in clinical practice and must be corrected.

ELECTROLYTE IMBALANCES			
HYPER ↑↑			
NA >145 MEQ/L	K >5.5 MEQ/L	CA >10.5 MEQ/L	MG >2.5 MEQ/L
<u>Cardiac:</u> Tachycardia, Increased BP <u>GI:</u> Thirst <u>GU:</u> Oliguria <u>Neuro:</u> Restlessness, anxiety <u>Skin:</u> Edema	<u>Cardiac:</u> T wave elevation, prolonged PR, Flat P wave, Wide QRS <u>GI:</u> Abdominal cramps <u>GU:</u> Oliguria <u>Neuro:</u> muscle cramps, weakness	<u>Cardiac:</u> Arrhythmias <u>GU:</u> Polyuria, kidney pain (kidney stones) <u>Musculoskeletal:</u> Bone pain, weakness	<u>Cardiac:</u> Hypotension, bradycardia, weak pulse <u>Resp:</u> Dyspnea, low RR <u>Neuro:</u> decreased loc, confusion <u>Musculoskeletal:</u> Muscle weakness, decreased reflexes
HYPO ↓↓			
NA <135 MEQ/L	K <3.5 MEQ/L	CA <8.5 MEQ/L	MG <1.5 MEQ/L
<u>Cardiac:</u> Tachycardia, thready pulse, hypotension <u>GI:</u> Nausea, Vomiting <u>Neuro:</u> Restlessness, headache dizziness, weakness, seizure	<u>Cardiac:</u> Hypotension, Arrhythmias, Flattened T-wave, ST depression <u>GI:</u> Nausea, Vomiting, decreased peristalsis <u>GU:</u> Polyuria <u>Neuro:</u> Dizziness, weakness, decreased reflexes	<u>Cardiac:</u> Arrhythmias, Bradycardia, Hypotension, weak pulse <u>Neuro:</u> Paresthesia, muscle spasms, seizures, +Trousseau signs, Chvostek signs <u>Resp:</u> Dyspnea, Laryngospasm	<u>Cardiac:</u> Arrhythmias, Tachycardia, High BP <u>Neuro:</u> Seizures, Delusions, Hallucinations <u>Neuromuscular:</u> Tetany, Chvostek signs, Positive Trousseau's
http://nursebossstore.com  nursebossessentials  nursebossstore			

Acid-Base Balance

- **Acid.** An acid is one type of compound that contains the hydrogen ion.
- **Base.** A base or alkali is a compound that contains the hydroxyl ion.
- **Salt.** Salt is a combination of a base and an acid and is created when the positive ions of a base replace the positive hydrogen ions of an acid.
- **Important salts.** The body contains several important salts like sodium chloride, potassium chloride, calcium chloride, calcium carbonate, calcium phosphate, and sodium phosphate.

Potential of Hydrogen

- **PH.** The symbol of pH refers to the potential or power of hydrogen ion concentration within the solution.

Buffers

- **Buffers.** A buffer is a chemical system set up to resist changes, particularly in hydrogen ion levels.

Causes of fluid and electrolyte imbalances:

- **Fluid retention.** Retention of sodium is associated with fluid retention.
- **Loss of sodium.** Excessive loss of sodium is associated with decreased volume of body fluid.
- **Trauma.** Trauma causes release of intracellular potassium which is extremely dangerous.
- **Loss of body fluids.** FVD results from loss of body fluids and occurs more rapidly when coupled with decreased fluid intake.

- **Fluid overload.** Fluid volume excess may be related to a simple fluid overload or diminished function of the homeostatic mechanisms responsible for regulating fluid balance.
- **Low or high electrolyte intake.** Diets low or excessive in electrolytes could also cause electrolyte imbalances.
- **Medications.** There are certain medications that could lead to electrolyte imbalances when taken against the physician's orders.

Nursing intervention:

- **Intake & Output.** The nurse should monitor for fluid I&O at least every 8 hours, or even hourly.
- **Daily weight.** Assess the patient's weight daily to measure any gains or losses.
- **Vital signs.** Vital signs should be closely monitored.
- **Physical exam.** A physical exam is needed to reinforce other data about a fluid or electrolyte imbalance.
- **Monitor turgor.** Skin and tongue turgor are indicators of the fluid status of the patient.
- **Urine concentration.** Obtain urine sample of the patient to check for urine concentration.
- **Oral and parenteral fluids.** Administer oral or parenteral fluids as indicated to correct the deficit.
- **Oral rehydration solutions.** These solutions provide fluid, glucose, and electrolytes in concentrations that are easily absorbed.
- **Central nervous system changes.** The nurse must be alert for central nervous system changes such as lethargy, seizures, confusion, and muscle twitching.
- **Diet.** The nurse must encourage intake of electrolytes that are deficient or restrict intake if the electrolyte levels are excessive.

Oxygenation

Introduction:

The main function of our respiratory system is to provide the body with a constant supply of oxygen and to remove carbon dioxide. To achieve these functions, muscles and structures of the thorax create the mechanical movement of air into and out of the lungs called ventilation. Gas exchange occurs at the alveolar level where blood is oxygenated and carbon dioxide is removed. Several respiratory conditions can affect a patient's ability to maintain adequate ventilation and respiration, and there are several medications used to enhance a patient's oxygenation status.

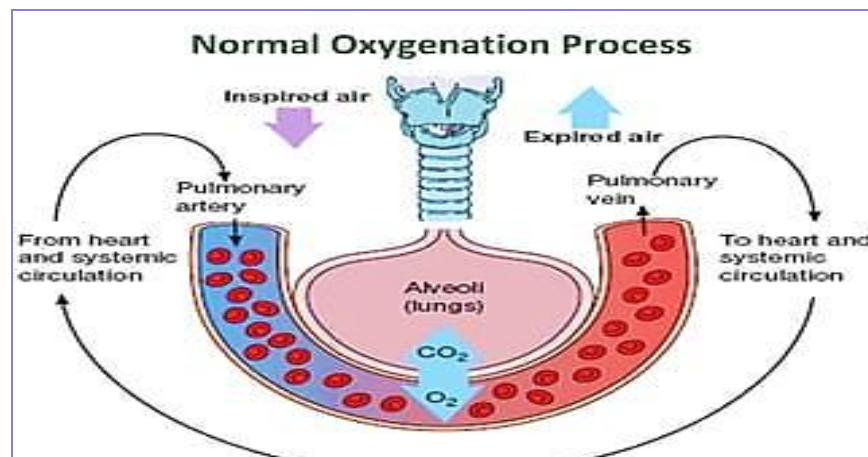
Definition:

Oxygen:

Is the most essential of all basic survival needs. Without oxygen circulating in the bloodstream, a person will die in a matter of minutes. Oxygen is provided to the cells by maintaining an open airway and adequate circulation.

Oxygenation:

It is process refers to how well the cells, tissues, and organs of the body are supplied with oxygen through the respiratory system functions.



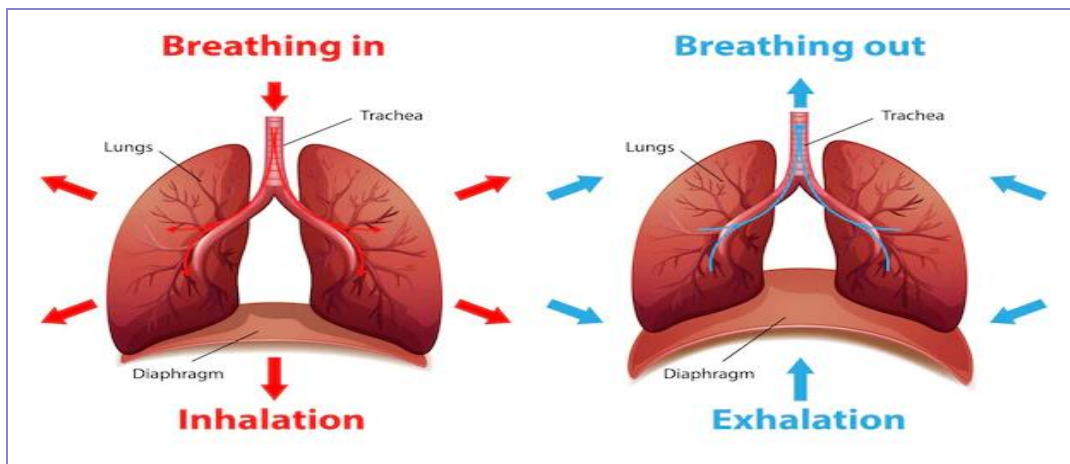
Respiration is the physiological process of taking in oxygen from the environment during inhalation, while getting rid of carbon dioxide during exhalation. It involves both ventilation and oxygenation.

Physiology of respiration:

While ventilation brings air into the lungs, it is oxygenation that helps ensure that the cells and body tissues receive enough oxygen molecules. The gaseous exchange occurs by diffusion in the alveoli. It depends on essentially three factors:

- The integrity of the airway system to transport air to and from the lungs.
- A properly functioning alveolar system in the lungs to oxygenate venous blood and to remove carbon dioxide from the blood.

Mechanism of ventilation:



1. **Inhalation (Inspiration)** Expansion of the chest cavity and lungs creates negative pressure inside the lungs, causing air to be drawn in through the nose or mouth and airways.
2. **Exhalation (Expiration)** When the diaphragm and intercostal muscles relax, exhalation allows the chest and lungs to return to their normal resting size.

Process of Oxygenation

- Ventilation: The process of moving gases into and out of the lungs.
- Perfusion: The ability of the cardiovascular system to pump oxygenated blood to the tissues and return deoxygenated blood to the lungs.
- Diffusion: Exchange of respiratory gases in the alveoli and capillaries.

Exchange of respiratory gases:

refers to, the oxygenation of blood and elimination of carbon dioxide in the lungs.

- 3. External Respiration** Alveolar capillary gas exchange (or external respiration) occurs in the alveoli of the lungs.
- 4. Internal Respiration** Capillary tissue gas exchange (or internal respiration) occurs in body organs and tissues.

Factors Affecting Oxygenation

- Physiological factors (Increased metabolic rate).
- Disease process (Anemia, Hypovolemia, Fever, Obesity & Dehydration and hemorrhage).
- Decreased inspired oxygen concentration
- Environmental factors (High altitudes & occupational pollutants).
- Factors affecting chest wall movement (Trauma)
- Pregnancy
- Musculoskeletal abnormalities
- Medications
- Lifestyle Risk Factors (Activity and exercise, Stress, smoking & substance abuse).

Common Respiratory Alterations

- **Hyperventilation:** ventilation in excess of that required; rate and depth of respirations increase.
- **Hypoventilation:** alveolar ventilation inadequate to meet the body's oxygen demand; respiratory rate and depth is low
- **Hypoxia:** Inadequate tissue oxygenation at the cellular level, late sign cyanosis
- **Hypoxemia:** Inadequate blood oxygenation.
- **Altered breathing patterns**
 1. Tachypnea (rapid rate)
 2. Bradypnea (abnormally slow rate)
 3. Apnea (cessation of breathing)
 4. Dyspnea & Orthopnea
- **Obstructed or partially obstructed airway:** As a result of the presence of foreign body, or accumulation of mucus or inflammatory exudates.

Nursing Interventions:

Airway Management:

- Administer humidified air or oxygen immediately
- Regulate fluid intake
- Monitor respiratory and oxygenation status
- Administer drug therapy (bronchodilators, corticosteroids)
- Auscultate lung sounds before and after treatments

Cough Enhancement

- Positioning for chest expansion
- Deep breathing, hold for 2 seconds, and cough 2-3 times.

Respiratory Monitoring:

- Rate, rhythm, depth, and effort (overall patterns)
- Monitor for increased restlessness, anxiety, and air hunger
- Note changes in oxygen saturation (SaO₂), arterial blood gases (ABG) values.

Anxiety Reduction

- Calming & reassuring attitudes
- Stay with patient
- Encourage slow breathing (pursed lips)

Teaching:

- Identify level of knowledge disease process & prescribed Medication
- Instruct on measure to prevent/minimize side effects of treatment
Evaluate patient's ability to self-administer medications
- Instruct patient on purpose, action, dosage, and duration of each medication
- Include family and significant others

Nutrition

Introduction:

Our bodies are built of and powered by solely what we eat and drink. Food is the source of all of the energy needed. Good nutrition is essential to wellness, and poor nutrition contributes to disease, so clients need accurate, current, and appropriate nutritional information.

Definition:

Nutrition: is a state of well-being achieved by eating the right food in every meal and the proper utilization of the nutrients by the body.

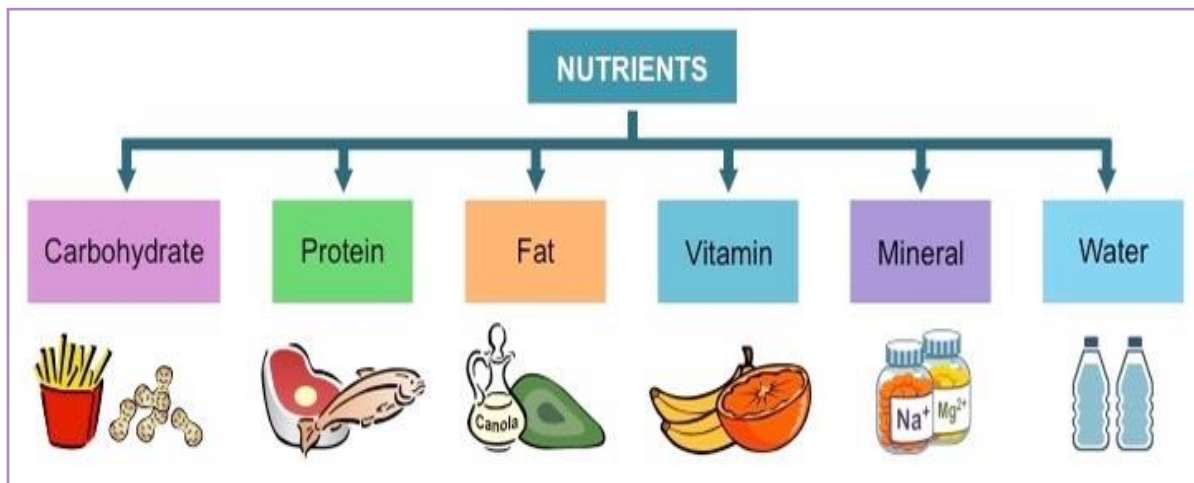
Proper nutrition is important because:

- It helps in the development of the brain, especially during the first years of the child's life.
- It speeds up the growth and development of the body including the formation of teeth and bones
- It helps fight infection and diseases
- It speeds up the recovery of a sick person
- It makes people happy and productive
- Proper nutrition is eating a balanced diet in every meal.

Nutrients:

It is the substances found in food which drive biological activity, and are essential for the human body.

- They are categorized as proteins, fats, carbohydrates (sugars, dietary fiber), vitamins, and minerals, and perform the following vital functions:
 - Building all parts of the body such as muscle, bone, teeth, and blood
 - Producing energy (power and heat)
 - Keeping the body in good working order.



- **Protein:** is the main constituent of the body, making up the muscles, internal organs, skin, and blood. they are essential for the growth, maintenance, and repair of body tissue.
- **Fats/carbohydrates (sugars): the body's energy (power and heat) source**
Fats (lipids) are the most calorie-dense nutrient, providing (9kcal per gram). Fats are composed of triglycerides and fatty acids. Although consuming too much fat can lead to obesity, small amounts can provide a highly efficient energy source.
- **Carbohydrates** can be broken down further into the two categories of sugars and dietary fiber. Sugars are the carbohydrates which can be used as an energy source to move the body (such as during exercise) and are stored in the liver and muscles as glycogen. Sugars are also the main source of energy for the brain.
- **Water.** Water is critical because cell function depends on a fluid environment. Water makes up 60% to 70% of total body weight.

- **Vitamins/minerals:** keeping the body in good working order Vitamins and minerals are not used as energy, but instead aid in breaking down and building up proteins, fats, and sugars, and are an essential nutrient for keeping the body healthy and in good working order.

Name of the Nutrient	Sources	Function	
Carbohydrates (energy giving food)	Rice, potato, wheat, sugar	Provides energy	
Fats (energy giving food)	Butter, ghee, milk, cheese	Gives more energy compared to carbohydrates	
Vitamins and minerals (protective food)	Fruits and vegetables	Required for normal growth and development	
Proteins (body building food)	Milk, eggs, meat, fish, soybean	Helps in building and repair of body	

Elimination

After ingesting food and fluids, our body eliminates waste products through the urinary system and the gastrointestinal system. Elimination is the expulsion of waste products from the body through the skin, lungs, kidneys and rectum.

Urinary elimination

Definition:

is the removal of waste products from the body through the urinary system(urine).

Physiology of urine elimination

- When the urine is collected in the bladder, desire to void is experienced due to the stimulation of stretch receptors.
- This sensation occurs when the bladder is filled with 250-450 ml of urine in adults and 50-200 ml in children.
- The stretch receptors transmit the message to the voiding reflex center in the spinal cord.
- If the time is appropriate the brain sends message through spinal cord causing stimulation so that the urine can be released from the bladder.
- If the time and place are appropriate the external sphincter and the urination process takes place.

Normal Urinary Function

- Normal urine output is 60mL/hr. or 1500mL/day; should remain 30 mL/hr to ensure continued normal kidney function
- Urine normally consists of 96% water
- Solutes found in urine include:
 - Organic solutes: urea, ammonia, uric acid and creatinine

- Inorganic solutes: sodium, potassium, chloride, sulfate, magnesium & phosphorus.

Factors Influencing Urinary Elimination

- Growth and Development
- Sociocultural Factors
- Psychological Factors
- Personal Habits
- Fluid Intake
- Pathological Conditions
- Surgical Procedures
- Medications
- Diagnostic Examinations

Common Urinary Elimination Problems

- **Anuria:** 24-hour urine output is less than 50 mL; synonyms are complete kidney shutdown or renal failure.
- **Dysuria:** Painful or difficult urination
- **Frequency:** Increased incidence of voiding
- **Glycosuria:** Presence of sugar in the urine
- **Nocturia:** Awakening at night to urinate
- **Oliguria:** Scanty or greatly diminished amount of urine voided in a given time; 24-hour urine output is less than 400 mL
- **Polyuria:** Excessive output of urine (diuresis)
- **Proteinuria:** Protein in the urine; indication of kidney disease
- **Pyuria:** Pus in the urine; urine appears cloudy
- **Suppression:** Stoppage of urine production; normally the adult kidneys produce urine continuously at the rate of 60 to 120 mL/h
- **Urgency:** Strong desire to void
- **Urinary incontinence:** Involuntary loss of urine

Bowel Elimination

Regular elimination of bowel waste products is essential for normal body functioning. Alterations in bowel elimination are often early signs or symptoms of problems within either the gastrointestinal (GI) tract or other body systems. Because bowel function depends on the balance of several factors, elimination patterns and habits vary among individuals. The gastrointestinal tract, also known as the alimentary tract or canal, extends from the mouth to the anus.

Definition:

Bowel Elimination (Defecation) is a natural process by which the soiled waste products of digestion (feces or stool) are eliminated from the bowel.

The Large Intestine

- Primary organ of bowel elimination
- Extends from the ileocecal valve to the anus

Functions

1. Completion of absorption of H₂O, Nutrients
2. Manufacture of some vitamins
3. Formation of feces
4. Expulsion of feces from the body

The Small and Large Intestines

Process of Peristalsis

- Peristalsis is under control of nervous system
- Contractions occur every 3 to 12 minutes
- Mass peristalsis sweeps occur 1 to 4 times each 24-hour period
- One-third to one-half of food waste is excreted in stool within 24 hours
- Peristaltic Movements in the Intestine – Colonic peristalsis is slow.

- Mass peristalsis is strong, few waves per day, stimulated by food in small intestine.

Factors that influence Bowel Elimination

- Age.
- Diet.
- Fluid Intake
- Physical Activity.
- Psychological Factors.
- Personal Habits.
- Position During Defecation.
- Pain.
- Pregnancy.
- Surgery and Anesthesia.
- Medications.
- Diagnostic Tests.

Common Bowel Elimination Problems

- **Constipation** – abnormal frequency of defecation and abnormal hardening of stools
- **Impaction** – accumulated mass of dry feces that cannot be expelled
- **Diarrhea** – increased frequency of bowel movements (more than 3 times a day) as well as liquid consistency and increased amount; accompanied by urgency, discomfort and possibly incontinence
- **Incontinence** – involuntary elimination of feces
- **Flatulence** – expulsion of gas from the rectum
- **Hemorrhoids** – dilated portions of veins in the anal canal causing itching and pain and bright red bleeding upon defecation.

Promoting Regular Bowel Habits:

- Timing -attend to urges promptly
- Positioning – have patient sit up, gravity aids in BM
- Privacy – close door & pull curtain
- Nutrition
- Exercise – abdominal muscles & thighs
- Abdominal settings
- Thigh strengthening

Rest & sleep

Physical and emotional health depends on adequate sleep and rest. Without the proper amount of rest and sleep, the patient's ability to concentrate, make judgments, promote healing, and participate in daily activity decreases. While caring for patients, you need to individualize an approach to sleep and rest based on the patient's personal sleep habits and pattern of sleep to provide effective sleep therapies.

Definition:

Rest: is a condition in which the body is inactive or engaging in mild activity, after which the person feels refreshed. A person at rest is calm, at ease, relaxed, and free of anxiety and stress.

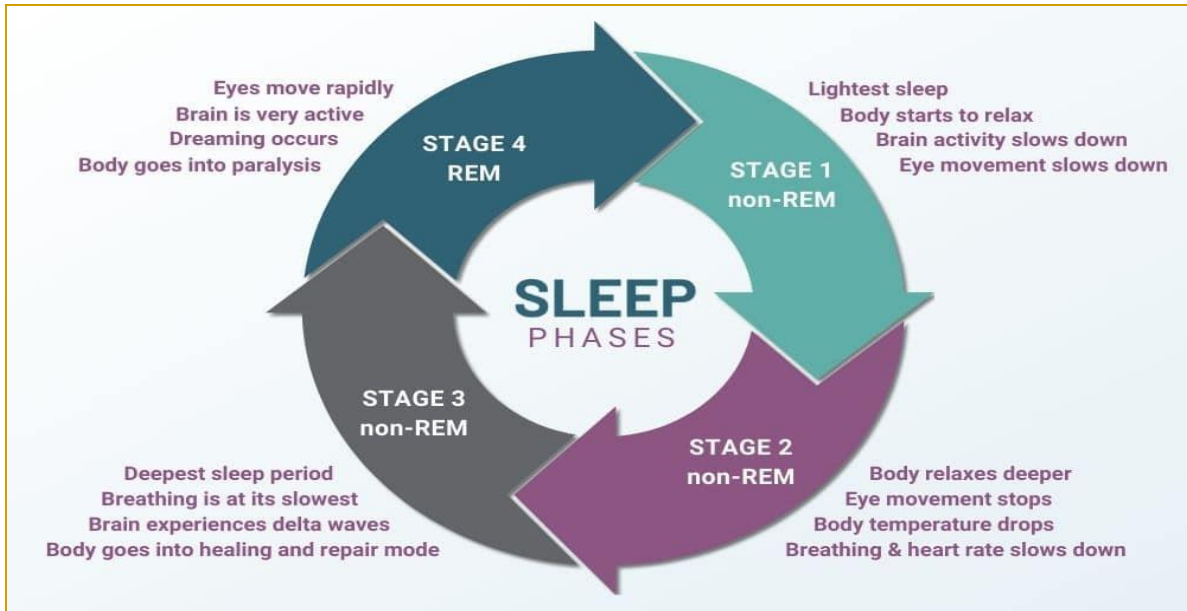
Sleep: is a state of rest accompanied by dynamic and regulated set of behavioral and physiological states which during many processes vital to health and well-being take place.

Sleep requirements and patterns:

- Eight hours of sleep a night has been the accepted standard.
- It varies according to age.
- Pattern of sleep periodicity appear to be learned.
- **For example**, most people learn to sleep at night and to be awake and work during the day. However, many night shift workers learn to sleep equally well during the day.

Physiology of Sleep

- Sleep is a cyclical physiological process that alternates with periods of wakefulness.

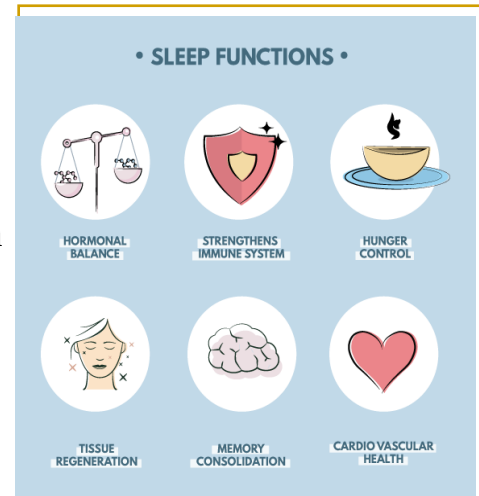


Functions of Sleep

1. Cleaning the brain of toxins
2. Physical restoration
3. Information processing & memorization
4. Mood regulation
5. Strengthening immune system

Factors affecting sleep

- Developmental considerations
- Physical activity
- Psychological stress
- Motivation
- Cultural implications
- Diet.
- Alcohol intake
- Caffeine-containing beverage



- Smoking
- Environmental factors
- Lifestyle
- Exercise
- Illness
- Medications

The more common sleep disorders are

- **Dyssomnias** that characterized by insomnia or excessive sleepiness.
- **Parasomnias** are patterns of waking behavior that appear during sleep such as somnambulism (sleepwalking), sleep talking, nocturnal enuresis (bedwetting during sleep).
- **Narcolepsy** is a condition characterized by uncontrollable desire to sleep.
- **Insomnia** is characterized by difficulty falling asleep, intermittent sleep, or early awakening from sleep.
- **Sleep apnea** refers to a period of no breathing between snoring intervals.
- **Sleep deprivation** refers to a decrease in the amount, quality of sleep.

Nursing Care for Patient with Sleep Disorders:

- Assessment of sleep pattern and take sleep history
- Preparing restful environment
- Promoting bedtime rituals
- Offering appropriate bedtime snacks and beverages
- Promoting relaxation
- Respecting normal sleep wake pattern
- Scheduling nursing care to avoid unnecessary disturbances
- Using medication to produce sleep
- Teaching about rest and sleep

Skin integrity

Introduction

The integumentary system consists of the skin, hair, nails, sweat glands, and the subcutaneous tissue below the skin. The skin is the largest organ of the body. The major functions of the skin include protection of the internal organs, unique identification of an individual, thermoregulation, metabolism of nutrients and metabolic waste products, and sensation.

Functions of the Skin

- Protection (first line of defense)
- Body temperature regulation
- Psychosocial
- Sensation
- Vitamin D production
- Immunologic
- Absorption
- Elimination

Factors Affecting Skin Integrity

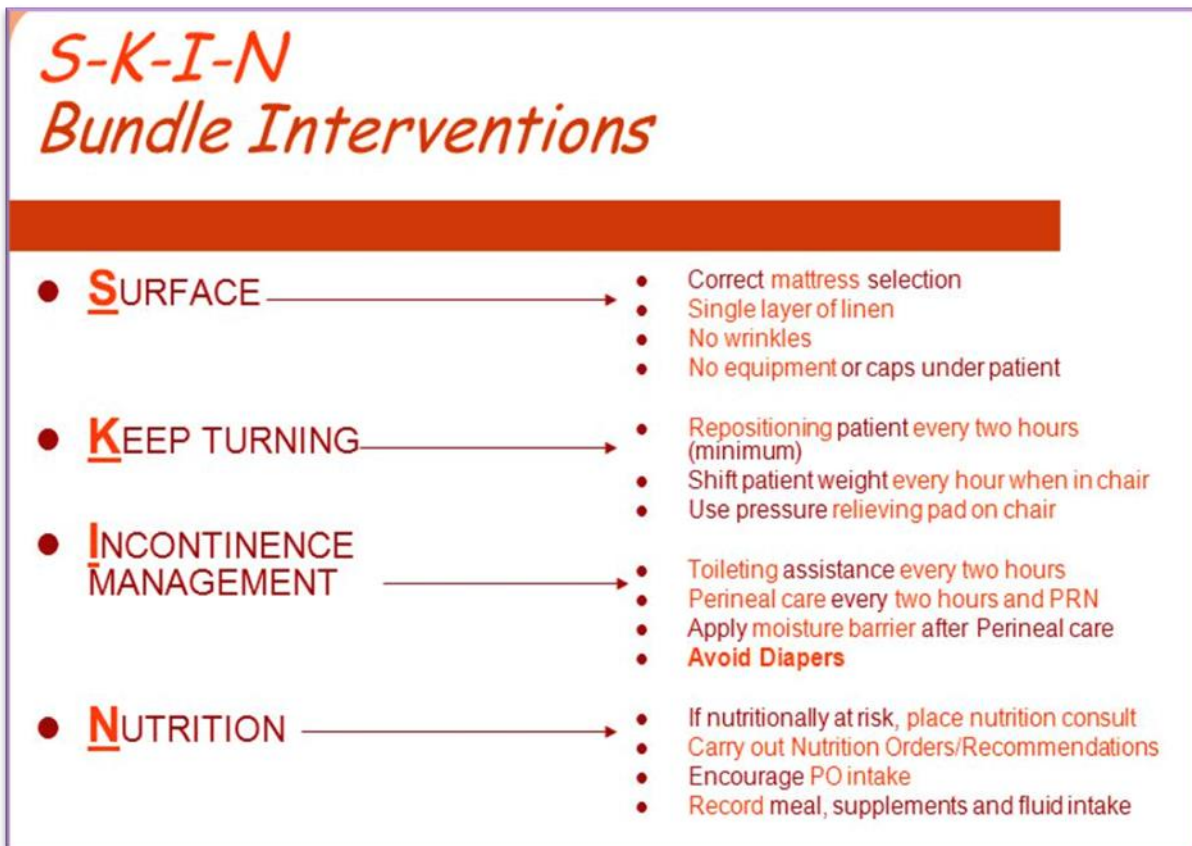
Basic principles related to integrity of the skin and mucous membranes include:

- Unbroken and healthy skin and mucous membranes serve as the first lines of defense against harmful agents.
- person's age, the number of underlying tissues, and illness conditions.
- Adequately nourished and hydrated body cells are resistant to injury.
- Adequate circulation is necessary to maintain cell life.

Care of the Skin:

- Limit use of tape on skin
- Careful skin assessments

- Changes in bathing
- Apply lotions
- Pressure-reducing interventions
- Adequate hydration



Patient Education

- Maintaining Intact Skin: adequate nutrition (fluids, protein, Vitamin B & C, iron) and proper skin hygiene.
- Appropriate positions for pressure relief
- Use of appropriate skin protection agents
- Positioning
- Report any reddened areas
- Promoting wound healing.

Sex concept

Introduction

Sexuality is a crucial part of a person's identity. Sex is central to who we are and to our emotional well-being. All people have the potential to positively experience and pleurably express their sexuality. Sexuality is always part of an individual professional nurse, as health care providers, focus on the holistic nature of care and have responsibility to provide effective sexual health care for their clients.

Definitions:

- **Sex:** 'One's biological characteristics: Anatomical, physiological and genetic (XX; XY)-as a female or as a male.
- **Gender:** Refers to what a person, society, or legal system defines as "female" or "male."
- **Gender role:** set of socially or culturally defined attitudes, behaviors, expectations, and responsibilities that is considered appropriate for women (feminine) and men (masculine).
- **Gender identity:** It is how one chooses to view oneself as a male or female in integration with others
- **Sexual health:** Integration of the physical, emotional, intellectual and social aspects of sexual being in ways that positively enhance personality. communication, pleasure, relationships and love Not merely the absence of disease, dysfunction or infirmity.

Reproductive system

Organs have dual functions:

- Produce reproductive cells
- Produce hormones responsible for sex characteristics
- Females - estrogen and progesterone
- Males – testosterone

Sexuality expressed by individuals of all ages

A way to show female or male qualities:

- Clothing styles and colors
- Hairstyles
- Hobbies and interests
- Sexual habits (continue into old age)
- Gestures

May be expressed by:

- Sexual intercourse
- Caressing, touching, holding hands.

Factors affecting sexuality

- Culture
- Religion
- Ethics
- Lifestyle
- Health state
- Age
- Stereotypes

Sexual Rights – WHO,

Sexual rights include the right of all persons, free of coercion, discrimination and violence, to the highest attainable standard of sexual health, including access to sexual and reproductive health care services seek, receive and impart information related to:

- sexuality education
- respect for bodily integrity
- choose their partner decide to be sexually active or not
- consensual sexual relations
- consensual marriage
- decides whether or not, and when, to have children
- pursue a satisfying, safe and pleasurable sexual life

Nursing intervention:

- Health promotion
- Illness prevention
- Provide support & counseling
- Encourage expressing their health goals and behaviors
- Instruction about personal hygiene, detecting & preventing diseases (STDs).
- Encourage healthy diet & exercises.
- stress management & well-being, healthy lifestyles, & avoiding risky behaviors

Activity

Introduction

Activities of daily living (ADLs) are essential and routine tasks that most young, healthy individuals can perform without assistance. The inability to accomplish essential activities of daily living may lead to unsafe conditions and poor quality of life. The healthcare team should be aware of the importance of assessing ADL in patients to help ensure that patients who require assistance are identified.

Definitions:

Physical activity (PA) is any movement produced by skeletal muscles that results in energy expenditure (e.g., occupational, sports, conditioning, and household activities).

Active living is an approach to life that values and includes physical activity in everyday living.

The activities of daily living (ADLs) are a term used to collectively describe fundamental skills required to independently care for oneself, such as eating, bathing, and mobility.

Importance of physical activity:

According to the American Heart Association (2018), physical activity:

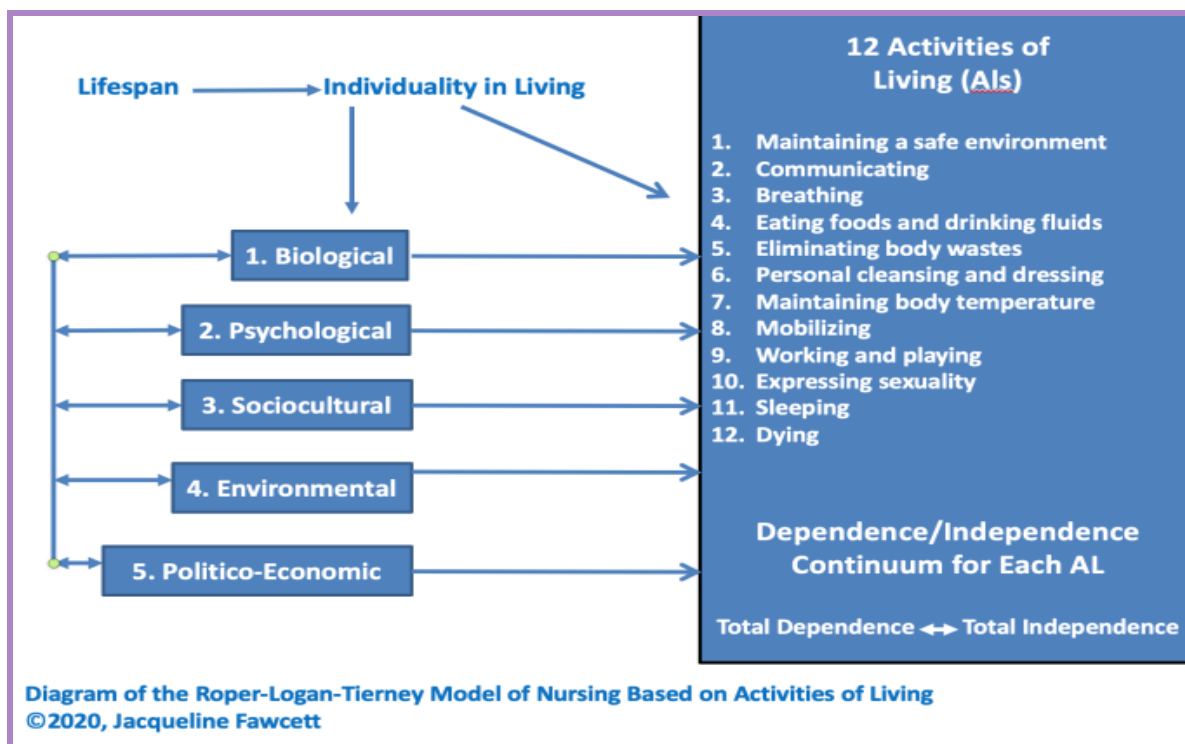
- Elevates mood and attitude
- Enables physical fitness
- helps when to quit smoking and stay tobacco free
- boost energy levels
- helps in the management of stress
- improve self-image and self confidence

Types of ADL

- The basic ADLs or physical ADLs are those skills required to manage one's basic physical needs, including personal hygiene or grooming, dressing, toileting, transferring or ambulating, and eating.
- The Instrumental Activities of Daily Living (IADLs) include more complex activities related to the ability to live independently in the community. This would include activities such as e.g., managing finances and medications, food preparation, housekeeping, laundry.

The 12 activities of living listed in the Roper-Logan-Tierney Model of Nursing are:

The model assesses the patient's level of independence in relation to the activities of living, which then helps the nurse and health care team to develop a nursing care plan based on the patient's individual abilities and levels of independence.



○ Maintaining a safe environment

This AL includes all those activities necessary to protect oneself and one's family, such as preventing accidents and infection at home or in the workplace, or, on a more global level, buying products that will not pollute the environment.

- **for example.** When a person requires nursing care his ability to maintain his own safety must be assessed in the context of his illness, treatment, dependence independence status, position on the life span, and a new environment or routine.



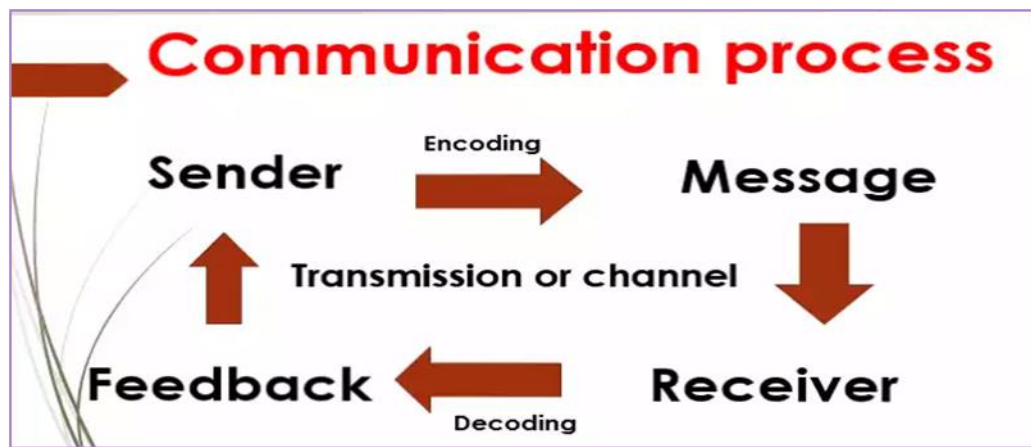
Nursing Intervention

- Use systematic risk assessments and communication (e.g., Falls risk, skin assessment)
- Assess surrounding for hazards, obstacles and remove it where possible.
- Safety education for all staff.
- Working within policies and practice standards.

○ Communication

Communication is an essential part of social interaction: it includes communication by all means - writing, reading, talking, and all forms of non-verbal communication. Problems with communicating may have physical causes, such as impairments of speech, hearing or sight; there may be psychological or intellectual reasons; or there may be cultural or language difficulties.

- **Verbal:** – language –accents, terminology appropriate for the persons
- **Nonverbal:** - tone, body language, touch culturally appropriate.



○ Breathing

Breathing is obviously the most vital activity, but when it is effortless and problem-free, it is one which every individual can carry out independently irrespective of age. Breathing and oxygen essential for life. Restricted or loss of oxygen for a short time can cause cell death

- **Factors affecting on breathing** are developmental age, health condition, and life style factors.

○ Eating and drinking

- This AL includes not only the physical acts of eating or drinking but also the activity of obtaining and preparing food.
- Problems may therefore include physical ones related to chewing, swallowing,
- self-feeding, the ability to reach the shops or to cook
- Eating and drinking may be a comforting activity, as well as an essential living activity, in times of unhappiness or stress

○ Eliminating

- An essential activity of living in which problems may cause enormous inconvenience, pain, embarrassment and misery.
- Problems may be caused by limited mobility, making it difficult to manage to get to the lavatory and to manage independently.
- Urinary tract infections may cause problems of acute discomfort and pain, while even a minor degree of incontinence will cause inconvenience and embarrassment, even leading to social withdrawal.

○ Washing and Dressing

Good personal hygiene is one of the best ways to protect yourself from getting gastro or infectious diseases such as COVID-19, colds and flu. Good personal hygiene involves keeping all parts of the external body clean and healthy. It is important for maintaining both physical and mental health.

- **Washing refers to** one's ability to wash their face or body in the bath or shower independently, as well as physically getting in and out of the shower.

- **Dressing refers to** the ability to choose which clothes to wear, putting them on, and managing one's self appearance.

○ **Controlling temperature**

The human body need a stable internal / core temperature irrespective of external environment for:

- Comfort.
- Function of body system.
- Core temperature peaks late afternoon, and is lowest during sleep.
- Extremities usually a little colder than the trunk of the body.

○ **Mobilization**

- **Mobility** the ability to move freely, easily, rhythmically and purposefully is an essential part of living.
- People must protect themselves from trauma and meet their basic needs.
- Assisting with mobility safety for patient and nurse.

○ **Working and playing**

- Associated with physical, psychological, social, emotional well being
- Health conditions particularly where there is some degree of physical or mental incapacitation limit a person to participate.

○ **10-Expressing sexuality**

- Expressing sexuality encompasses more than sex and sexual activity. It relates also to how we see ourselves and our bodies in relation to each other and how we behave in society
- This can include intimate information – may be difficult to discuss may not be culturally appropriate.

11-Sleeping

On average, people spend one third of their lives asleep, the functions of sleep are usually characterized as being protective and restorative.

The sleep accomplishes the following functions:

- Restores physical well-being
- Relieves stress and anxiety
- Restores the ability to cope
- Concentrate on activities of daily living

12-Death and dying

- **Death** is inevitable part of life- marks the end of physical life
- It is essential however that nurses that aware of their own believes, values, ideas and actively support own well-being.

Factors influencing activities of living

- **Biological:** The biological factor addresses the impact of the overall health, of current injury and illness, and the scope of the patient's anatomy and physiology.
- **Psychological:** The psychological factor addresses the impact of emotion, cognition, spiritual beliefs, and the ability to understand.
- **Socio-cultural:** includes expectations and values based on class, status, and culture within the sociocultural factor relates to the beliefs, expectations, and values held by the individual patient for him or herself, as well as by others pertaining to independence in and ability to carry out the activities of daily living.

E. Environmental: not only the impact of the environment on the activities of daily living, but also the impact of the individual's ALs on the environment.

F. Politico-economic: is the impact of government, politics and the economy on ALs. Issues such as funding, government policies and programs, state of war or violent conflict, availability and access to benefits, political reforms and government targets, interest rates and availability of funding (both public and private) all are considered under this factor.

Nursing intervention of ADL:

Activities of daily living should be viewed “As a cognitive approach to the assessment and care of the patient, The patient should be assessed on admission, and his or her dependence and independence should be reviewed throughout the care plan and evaluation.

Activity	Assessment
Maintaining safe environment	Physiologic data – T, P, R, BP ▪ Vision and hearing ▪ Mobility vs immobility ▪ Tissue viability assessment ▪ Falls assessment
Communication	1. Language spoken- interpreter? 2. Mental capacity 3. Hearing ability 4. Speech ability 5. Communication aids

Breathing	▪ Clear airway ▪ Observe breathing rate, depth and effort ▪ Pain related to breathing ▪ Smoking history
Eating & Drinking	• Hydration • Nutrition • Swallowing and chewing • Patterns of eating and drinking • Dietary requirements/ religious requirements • Weight and BMI • Allergies
Elimination	• Elimination habits • Continence • Colostomy or ileostomy
Personal cleansing and dressing	▪ Hygiene needs ▪ Assistance with dressing/ undressing ▪ Usual hygiene practices
Controlling Body Temperature	▪ Take temperature ▪ Ability to maintain own temperature
Mobilizing	▪ Normal gait ▪ Aids used for mobilizing ▪ Moving and handling assessment
Working & playing	▪ Employment ▪ Hobbies ▪ Sports
Expressing Sexuality	▪ Problems related to sexual function
Sleeping	▪ Normal sleep pattern ▪ Medication ▪ Activities to promote sleep
Dying	• Cultural and religious beliefs • Family/ friends • Next of kin

Death and dying

Definition

- Death is defined as "The irreversible cessation of all vital functions especially as indicated by permanent stoppage of the heart, respiration, and higher brain function.
- " Dying means "approaching death".

Physical signs that death is near include:

- A sudden burst of energy
- Mottled and blotchy skin, especially on the hands, feet and knees.
- Blood pressure decreases
- They cannot swallow
- Less urine
- Restlessness
- Difficult breathing
- Congested lungs

Care of dying

- To provide effective care of dying individual nurse must have reconciled his or her own feelings about death and must understand the phases of grieving & dying and should be able to recognize their manifestations.
- Every person has the right to die with dignity.
- Nurse must understand the influence of distinguished death and the profound effect it has on the family and those close to the person who has died.

- Ensuring a good death for all is therefore a major challenge not only for healthcare professionals but also for society.
- Assessing physical needs, psychological needs & spiritual needs
- Explaining the client's condition and treatment
- Maintaining good communication
- Promoting self-care & self esteem
- Allowing family members to assist in care.

Students' self-assessment

Which of the following levels of basic human needs is most basic?

- a. Physiologic
- b. Safety & Security
- c. Love & belonging
- d. Self-actualization

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Patient X is diagnosed with constipation. As a knowledgeable nurse, which nursing intervention is appropriate for maintaining normal bowel function?

- a. Assessing dietary intake
- b. Decreasing fluid intake
- c. Providing limited physical activity
- d. Turning, coughing, and deep breathing

Which patient is at most risk for fluid volume deficient?

- a. A patient who has been vomiting and having diarrhea for 2 days
- b. A patient with continuous nasogastric suction.
- c. A patient with an abdominal wound vac at intermittent suction.
- d. All of the above are correct.

The nurse goes to assess a new patient and finds him lying supine in bed. The patient tells the nurse that he feels short of breath. Which nursing action should the nurse perform first?

- a. Raise the head of the bed to 45 degrees.
- b. Take his oxygen saturation with a pulse oximeter.
- c. Take his blood pressure and respiratory rate.
- d. Notify the health care provider of his shortness of breath

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Chapter Five

Admission and discharge from hospital

Learning objectives:

At the end the students will be able to:

- Identify admission process
- Illustrate methods of patient admission
- Identify admission specialty
- List important facts on admission
- List patient rights and responsibilities
- Identify patient transfer
- Determine types of transfer
- Determine types of referrals
- Identify patient discharge
- Determine the joint commission recommendation for discharge planning process.
- list elements of a written discharge summary form

Introduction

The coordination of resources and planning a patient's care from admission to discharge or from one level of care to the next is a key role of a nurse. Discharge planning begins at the time of admission to a hospital, or even earlier. Patients and families should be partners in care, sharing in the process of decision making. Furthermore, the admission assessment must incorporate patients' cultural beliefs and practices so can provide a patient-centered approach to care.

Admission process

Definition:

Admission is defined as allowing a patient to stay in hospital for observation, investigation, treatment and care.

Admission is the entry of a patient into a hospital /ward for therapeutic /diagnostic purposes.

Purposes of admission

- For diagnostic investigations to be done
- For treatments which may be medical or surgical
- For observation.

Patient Psychosocial Responses on Admission

- Anxiety and fear
- Decisional conflict
- Situational low self-esteem
- Powerlessness
- Social Isolation – Loneliness
- Decreased Privacy Loss of Identity

Types of admission

1. Emergency admission

In this, patients are admitted in acute conditions requiring immediate treatment. **E.g.**, Patient with RTA, Poisoning, burns and respiratory emergency.

2. Routine admission

In this, patients are admitted for investigation, diagnostic and medical or surgical treatment. Treatment is given according to patients' problem. **E.g.**, Patient with hypertension, diabetes mellitus etc.

Admission process:

- Patient Identification: The team uses a unique client identifier such as a given number, to link the record to the client.
- Detect payment method
- Choose room category
- Granting consent
- Admission officer inform in-charge nurse of the unit about the new admission
- Submit admission on the system
- Patient relation coordinator or nurse accompany patient to his /her room

Methods of patient's admission:

1-Direct admission from out-patient clinics by outpatient department (OPD) physician:

- Patient admitted to regular room
- Medical secretary guides patient from admission office to his room
- Staff nurse of the ward receives patient after room orientation.

2-Admission from emergency room by ER physician:

- Perform ER medical assessment that includes complete history, examinations, investigations and emergency medications
- Decision upon for need for admissions:
 - Consultations then admission to general ward (medical or surgical), intensive care units (ICUs) or isolation
 - Discharge to home or transfer to another hospital
 - In registration process in outpatient settings (OPD, ER, Lab , Radiology...), the following data are MUST to be recorded:
 - Patient's full name
 - Gender
 - Date of Birth
 - National ID Number/Passport number

Admission specialty Examiners:

- Non-surgical patients less than 12 years examined by pediatric
- Pregnant women examined by obstetrics and gynecology
- Poly Trauma adult patient needs two specialties to manage is prioritized needs according to trauma guidelines
- Poly Trauma adult patient needs more than two specialties to manage is examined by general surgery.

Important facts on admission:

- Triage in ER is done by triage nurse in triage room by using triage
- Maximum time for patients to stay in ER is two hours, then patients should be admitted, transferred or discharged
- Patient should not be admitted, transferred, or referred before diagnostic/urgent investigation's results are available, unless patient needs

immediate specific care and/or stabilization.

- Admission to critical/specialized care units should comply with the admission **criteria** of these units as ICU, CCU, neonatal intensive care unit (NICU) and pediatric intensive care unit (PICU)
- The OPD nurse must inform the physician immediately if patient has any of the following:
 1. Heart rate over 140/min or less than 50/min
 2. Respiratory rate over 25/min or less than 12/min
 3. Systolic blood pressure greater than 180 mmHg or less than 90 mmHg
 4. Disturbed conscious level
 5. Patient is presented with severe pain or chest pain
 6. Patient is presented with signs of communicable diseases.
- In cases of unavailable beds in a unit assigned for patient admission, **Uniformity of care** will be achieved by admitting patients in another unit providing the same level of needed care,
- **Example:** if patient is recommended to be admitted to ICU and no beds are available in ICU, the patient can be admitted to another critical care area as cardiac care unit (CCU).
- If there are no available beds in all critical areas, the same care will be provided to the patient in ER till a proper bed in another hospital is found.
- If patient is having acute physical problems, postpone routine admission procedures and nursing history until his or her needs are met. Complete a focused assessment at this point.

Patient rights and responsibilities

1- Patient rights

1. identify the place of phone, call bell, time of relative visit, and toilet.
2. Questions answered so patient understand his condition.
3. Participates in decisions of care.
4. Give informed consent to treatments.
5. know his care providers.
6. Participate in end-of-life decisions.

2- Patient responsibilities

- 1- Respect the autonomy of doctors and nurses.
- 2- Treat doctors and nurses with respect.
- 3- Keep hospital building clean.
- 4- Follow doctor instruction diligently.
- 5- Make the payment of treatment.

Nursing responsibilities in admission:

- Preparing the client's room
- Greeting the patient, meet and receive the patient.
- Collecting insurance information
- Verify the patient data, by checking the record sheet, chart.
- Take care of patient's valuables.
- Ensure patients sign consent form for treatment.
- Ask the patient to change clothes into hospital gown if necessary
- Recording vital signs & obtaining a medical history
- Orient patient and relatives regarding hospital policies
- Coordinating with physicians
- Preparing patients for operations

- Assist patient to the treatment area
- Starting basic treatment
- Performing discharging procedures
- Educating patients and their families

Transfer Process

Definition:

Transfer: “discharging a client from one unit or agency and admitting him or her to another without going home”

➤ **Transfers are used when there is a need to:**

- Facilitate more specialized care in a life-threatening situation
- Reduce health care costs
- Provide less intensive nursing care

The goal of patient transfer:

- To continue health care
 - To avoid therapeutic interruption or omissions that may hinder progress toward recovery.
- The physician writes order of transfer in the physician order form
- Physician determines the transfer needs and documents any special considerations of transfer in the medical records.

Types of transfer:

1-Internal transfer:

When patient is transferred between the same specialty or between different specialties

2-External transfer:

When the responsibility of the care transferred to another consultant outside the hospital.

3-Intra-hospital transfer

When the patient is transferred physically to another unit (Temporary-Permanent)

- Temporary: from ward to Cath lab.
- Permanent from ward to ward or ward to ICU

Referral Process

Definition:

A referral: process of sending someone to another person or agency for special services referrals generally are made to private practitioners or community agencies.

Types of referrals:

1-Internal:

Decision is taken for referral to another physician inside the hospital for

- Following recommendation of treating physician
- Patient and family preference
- Company agent request.

2-External:

Decision is taken for referral to another physician outside the hospital

- Patient is assessed and decision is taken for referral to another physician based on treating physician recommendation or patient and family or company agent request.
- This arrangement could be verbally via phone, medical report.

Indication for referral to another hospital:

- Patient 's need for specialized care not available in hospital
- Non availability of hospital beds
- Preference of patient and family
- Illegibility for treatment

Patient's discharge

Discharge occurs when a patient leaves the hospital after a period of treatment to his or her home; it normally done at the discretion of the medical team when patient is fit or his condition is stable or upon patient's own request. It is important that patients and relative have a prior knowledge of the intended discharge.

Discharge planning:

It is a process that facilitate the transition of the client from the health care institution to the most independent level of care, home or another health facility.

Goal of discharge planning is:

- To provide the most appropriate level and quality of care throughout all stages of the client illness.
- To ensure adequate continuity of care.

The Joint Commission Recommendations for Discharge Planning Process:

- Engage patients and families actively in discharge planning and instruction that meets patient needs
- Instruction may involve use of pictures, diagrams, or models to illustrate instruction.
- Identify follow-up providers who can meet unique patient needs.
- Create a list of follow-up providers that offer services and accommodations that meet patient's communication, cultural, religious, mobility, and other needs.

Elements of a Written Discharge Summary Form:

❖ Discharge instructions include:

- Medication
- Diet
- Special care and education related to newly diagnosis at home
- How to manage discomfort or any complain or expected side effects
- Discuss the discharge plan with patient or family
- Call to other specialty involved in plan of care

❖ Discharge summary includes:

- Reason for admission
- Diagnosis
- Medication
- Patient condition on discharge
- Significant findings
- Procedures performed
- Other treatments

Students' self-assessment

A patient is admitted for observation and various diagnostic tests. What is the initial nursing action in the admission process?

1. The nurse introduces self and roommates.
2. The nurse measures vital signs.
3. The nurse helps the patient get undressed and into bed.
4. The nurse notifies the physician.

A patient has been transferred out of the ICU to a medical unit. A nurse has been assigned to complete the transfer. Which type of transfer is this?

1. Patient-initiated transfer
2. Interagency transfer
3. Business office transfer
4. Intra-agency transfer

A patient is being transferred to the surgical unit from the recovery room after extensive surgery as the result of trauma from an automobile accident.

What concept should the nurse remember when transferring this patient?

1. The patient is a human who deserves dignity, courtesy, and respect.
2. The patient is ill and unable to make decisions or give accurate information.
3. The nurse knows best and should tell the patient what to do.
4. Families get in the way and should be encouraged not to get involved in the patient's care.

A patient has been recently diagnosed and hospitalized for type 1 diabetes. The multidisciplinary health care team has been preparing her for discharge. What is the primary purpose of discharge planning?

1. Verifying prescribed medications
2. Providing accurate medical treatment
3. Supplying ongoing patient education
4. Ensuring continuity of care

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Chapter Six

First Aid

Learning objectives:

At the end of this chapter the student will be able to:

- Define first aid and first aid provider.
- List the aims of first aid.
- Recognize the responsibility of a first aider.

1- Medical Emergency:

- Demonstrate first aid for breathing problem.
- Distinguish between basic life support (BLS) in adult, news in a Child, and Infant.
- Respond to choking in adult or child, pregnant women or large adult or child and infant.
- Explain how to care for someone who has fainted.
- Apply first aid for diabetic emergencies.

2- Injury emergency:

- Explain how to deal with each condition of external bleeding.
- Describe the signs and first aid for internal bleeding.
- Demonstrate first aid for head, neck and spine injuries.
- Demonstrate first aid for burns and electrical injuries.

3- Environment emergency:

- Executing the first aid for all bites and stings.
- Implementing first aid for heat-related emergencies.
- Implementing first aid for cold-related emergencies

Introduction

First aid refers to medical attention that is usually administered immediately after the injury occurs and at the location where it occurred. It often consists of a one-time, short-term treatment and requires little technology or training to administer.

Definition

First aid is emergency care provided for injury or sudden illness before emergency medical treatment is available.

Aims of a first aid (3 PS):

- » Preserve life.
- » Prevent injury/ condition from worsening.
- » Promote recovery.

The first-aid provider:

Is a person who takes charge of an emergency scene and give first aid.

Responsibilities of a First Aider:

- **Assess the situation quickly and calmly.**
- **Comfort, reassure, stay calm and take charge.** You should introduce yourself to the casualty, explain what's happening and why, and explain what you're going to do before you do it.
- **Protect yourself and the casualty from any danger.**
- **Prevent infection between you and them.**
- **Assess the casualty.** If there are multiple casualties, you must help those with life-threatening injuries or conditions first.
- **Give first aid treatment,** such as CPR or applying pressure to bleeding wounds.
- **Arrange for appropriate help.** Call for an ambulance if it's serious, or take/send them to a hospital if it's serious.

Emergency condition:

- **Medical emergencies**
- **Injury emergencies**
- **Environment emergencies**

- **Medical emergencies.**

➤ Breathing problems:

Definition:

Breathing problems involves a sensation of difficult or uncomfortable breathing or a feeling of not getting enough air.

Causes of breathing problems:

Breathing problems can arise from underlying lung diseases such as asthma or emphysema, as well as from illnesses such as pneumonia.

Signs of breathing problems:

1. Is breathing very fast or very slowly.
2. Is having trouble with every breath.
3. Has noisy breathing- hear a sound or whistle as the air enters or leaves the lungs.
4. Can only make sounds or speak no more than a few words at a time in between breaths although the person is trying to say more.

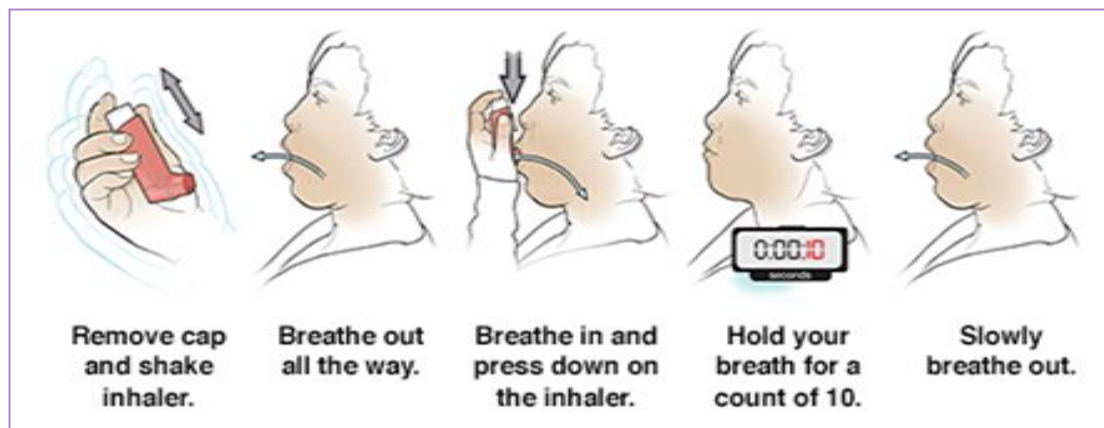
First aid for helping someone with breathing problem:

- Make sure the scene is safe.
- Ask the person if he need help. If he does, ask if he has medicine (such as: an inhaler).
- If he has medicine, get it for him. Then, assemble and help him use the inhaler.

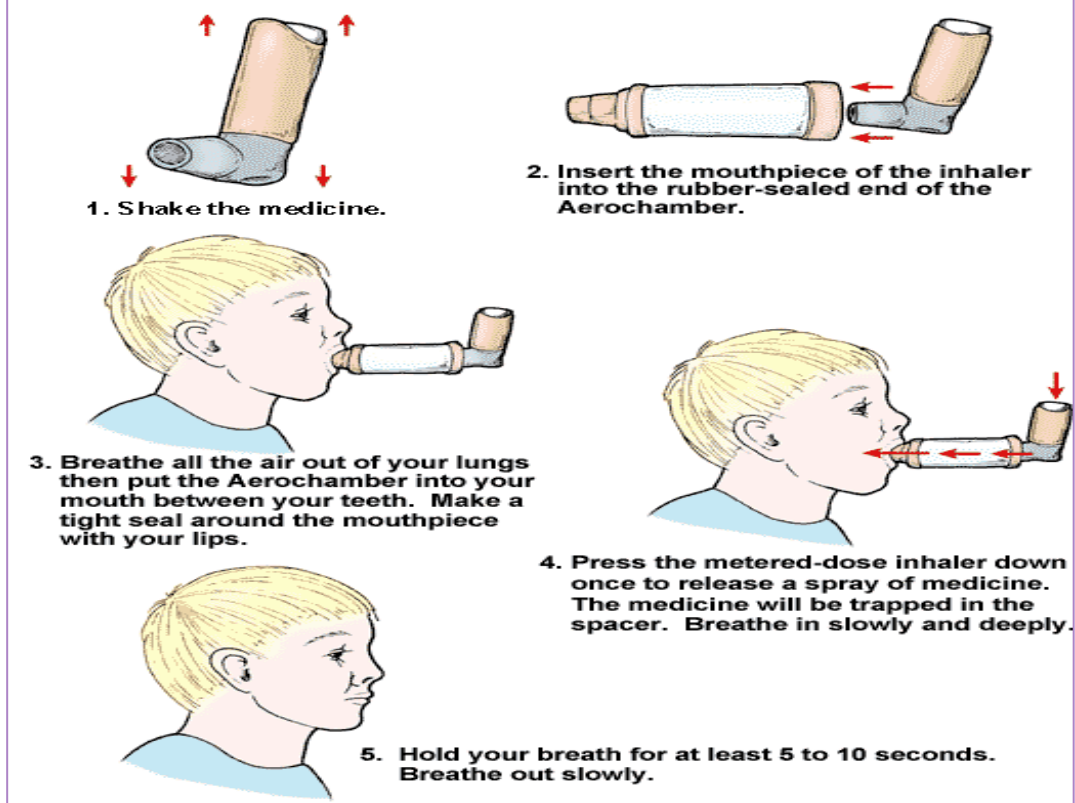
- Phone to local emergency if:
 - The person has no medicine.
 - The person does not get better after using his medicine.
 - The person`s breathing gets worse.
 - The person has trouble speaking.
 - The person becomes unresponsive.

Technique for using an inhaler:

- Shake the inhaler canister.
- Place the opening of the inhaler into the spacer if available.
- Instruct the person to fully exhale.
- Place the spacer or inhaler into their mouth.
- Simultaneously have the person inhale slowly and deeply while pressing down on the top of the inhaler canister.
- Instruct the person to hold their breath for up to 10 seconds if possible.
- Be prepared to repeat if respiratory problems persist.
- Stay with the person until the symptoms improve or until emergency response arrives.



How to Use a Metered-Dose Inhaler with an Aerochamber (Spacer)

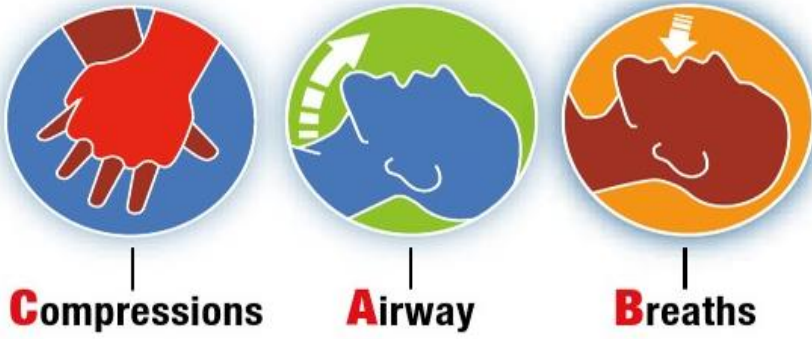


- In case of person is unresponsive, shout for help, check pulse, and Check breathing in the same time of checking pulse by monitoring chest raise (not less than 5 sec and not more than 10 sec).

Cardiopulmonary resuscitation

Definition of cardiopulmonary resuscitation (CPR) is an emergency procedure or is a lifesaving technique that can help save a person's life if their breathing or heart stops

C-A-B



Compressions **Airway** **Breaths**

Critical Concepts

High-quality CPR improves a victim's chances of survival. The critical characteristics of high-quality CPR include the following:

- **Start compressions within 10 seconds** of recognition of cardiac arrest.
- **Push hard, push fast:** Compress at a rate of 100 to 120/min with a depth of
 - At least 2 inches (5 cm) for adults
 - At least one third the depth of the chest, about 2 inches (5 cm), for children
 - At least one third the depth of the chest, about 1½ inches (4 cm), for infants
- **Allow complete chest recoil** after each compression.
- **Minimize interruptions** in compressions (try to limit interruptions to less than 10 seconds).
- **Give effective breaths** that make the chest rise.
- **Avoid excessive ventilation.**

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Difference between CPR in Adult, news in a Child, and Infant Adult

	Adult	Child	Infant
Compression to Ventilation Ratio	30 Compressions to 2 breaths in case of one or two rescuers.	30 Compressions to 2 breaths in case of One rescuer, and 15 Compressions to 2 breaths in case of Two rescuers.	30 compressions to 2 breaths in case of one rescuer, and 15 compressions to 2 breaths in case of two rescuers.
Compression Depth	At least 5 cm	About 5 cm	About 4 cm
Hands Position	In the center over the lower half of the sternum by using two hands	In the center over the lower half of the sternum by using one or two hands	In the center of the chest under the nipple line using two fingers of one hand or two thumb encircling technique with one or two rescuers. Also, can use the heel of one hand if you can't reach to the suitable depth.

Heartsaver®
Adult CPR AED



American Heart Association®
life is why™



Tap and shout.



Shout for help. Send someone to phone 9-1-1 and get an AED.



Look for no breathing or only gasping.



Push hard and fast at a rate of 100 to 120 compressions per minute.



Open the airway and give 2 breaths.



Repeat sets of 30 compressions and 2 breaths.



When the AED arrives, turn it on and follow the prompts.

Heartsaver®
Child CPR AED



American Heart Association®
Learn and Live



Tap and shout



Yell for help. Send someone to phone 911 and get an AED



Look for no breathing or only gasping



Push hard and fast. Give 30 compressions



Open the airway and give 2 breaths



Repeat sets of 30 compressions and 2 breaths



If you are alone after 5 sets of 30 compressions and 2 breaths, phone 911, and then resume sets of 30:2



When the AED arrives, turn it ON and follow the prompts

Heartsaver®
Infant CPR



American Heart Association®
Learn and Live



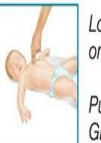
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Repeat sets of 30 compressions and 2 breaths



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Chocking

Definition:

Chocking is when food or another object gets stuck in the airway in the throat; the object can block the airway and stop air from getting to the lungs.

Causes:

1- **In adults**, chocking is often caused by food. **In children** can be caused by food or another object.

Sign of chocking:

- 1- If someone is chocking, he might use the chocking sign, which is holding the neck with one or both hands.
- 2- Coughing or gagging.
- 3- Wheezing chest.
- 4- Difficulty in breathing, cyanosis.
- 5- Decrease level of consciousness.



First aid to help a chocking adult or child with airway block:

- If you think someone is chocking, ask, “Are you chocking? Can I help you?”
- If the person nods yes, tell him you are going to help.

- Stand firmly or kneel behind the person (depending on your size and the size of the person choking). Wrap your arms around the person's waist so that your fists are in front.
- Make a fist with one hand.
- Put the thumb side of your fist slightly above the belly button and well below the breastbone.
- Grasp the fist with your other hand and give quick upward thrusts into the abdomen.
- Give thrusts until the object is forced out and the person can breathe, cough, or speak.



First aid to help a choking pregnant women or large adult or child with airway block:

- If you can't wrap your arms fully around the waist, give thrust on the chest (chest thrusts) instead of on the abdomen.
- Put your arms under the armpits and your hands on the lower half of the breastbone.
- Pull straight back to give chest thrusts.



First aid to help a choking infant with airway block:

- Hold the infant face down on your forearm. Support the infant's head and jaw with your hand.
- Give up to 5 back slaps with the heel of your other hand, between the infant's shoulder blades.
- If the object does not come out after 5 back slaps, turn the infant onto his back, supporting the head.
- Give up 5 chest thrusts, using 2 fingers of your other hands to push on the chest in the same place you push during CPR.
- Repeat giving 5 back slaps and 5 chest thrusts until the infant can breathe, cough, or cry.



Remember:

If you can't remove the object blocking the airway, the person will become unresponsive. Always give CPR to anyone who is unresponsive and not breathing normally or only gasping.

Unresponsive		provide
+	=	
No breathing or only gasping		CPR

Fainting / Syncope

Definition:

It is a sudden and temporary loss of consciousness and muscle strength characterized by a fast onset, short duration, and spontaneous recovery. It is caused by a decrease in blood flow to the brain.

Causes of Fainting / Syncope:

- Standing up too quickly – this could be a sign of low blood pressure
- Not eating or drinking enough
- Being too hot
- Being very upset, angry or in severe pain
- Heart problems
- Taking drugs or drinking too much alcohol

Symptoms of Fainting / Syncope:

Fainting usually happens suddenly. Symptoms can include:

- Dizziness
- Cold skin and sweating
- Feeling warm or hot
- Feeling sick
- Changes to your vision

First aid to help a person who has fainted and is responsive:

- Ask the person to continue to lie flat on the floor until he can sit up and feels normal.
- If the person fell, look for the injuries caused by the fall.
- Phone your local emergency number.
- If the person becomes unresponsive, give CPR.

Hypoglycemia:

Hypoglycemia occurs when the insulin level is too high relative to the level of sugar in the blood. The small amount of sugar in the blood gets used up quickly.

Causes of Hyperglycemia:

- Misses a meal or snack.
- Exercises more than usual.
- Vomits, or takes too much medication.

Symptoms of hyperglycemia:

- Rapid onset of confusion, irritability, or combativeness.
- Hunger. - Rapid pulse rate.
- Shallow breathing. - Pale, clammy skin.
- Loss of co-ordination, tremors, slurred speech.
- Generalized weakness. - Headache, dizziness.
- Seizures.

First aid to take for a responsive person who has hypoglycemia:

- If the person can sit up and swallow
 - Ask the person to eat or drink something with sugar that can rapidly restore blood glucose levels. These items include glucose tablets, orange juice or soft candy.
- If the person can't sit up or swallow, don't force her
 - Have the person sit quietly or lie down.
 - If the person does not improve within 15 minutes, phone or have someone phone your local emergency number.

Injury emergencies

» **External bleeding:** Most bleeding can be stopped with pressure.

First aid to control bleeding:

- Make sure the scene is safe.
- Send someone to get the first aid kit.
- Put on personal protective equipment (PPE).
- Apply dressing from the first aid kit. Put direct pressure on the dressings over the bleeding areas. Use the flat part of your fingers or the palm of your hand.
- If the bleeding doesn't stop, you will need to add a second dressing and press harder. Don't remove a dressing once it's in place because this could cause the wound to bleed more. Keep pressure on the wound until it stops bleeding.
- Once the bleeding has stopped or if you can't keep pressure on the wound, wrap a bandage firmly over the dressings to hold them in place.
- For minor cuts, wash the area with soap and water. Then, apply a dressing to the wound.

➤ Bleeding from the Nose (Epistaxis):

Definition:

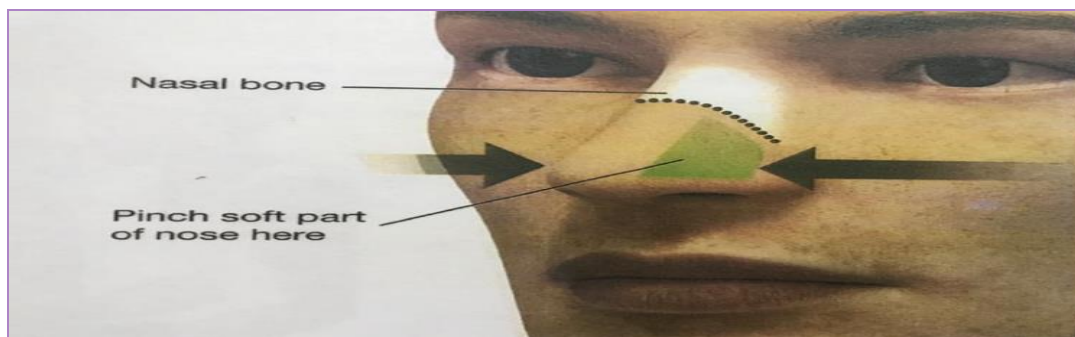
Epistaxis" is the medical term for a nosebleed. A nosebleed, meaning a loss of blood from the tissue that lines the inside of your nose, can occur in one or both nostrils.

Causes of Nosebleed:

- Colds/ sneezing and blocked nose such as with hay fever.
 - Dry nasal mucosa.
 - High blood pressure.
 - Cocaine use.
 - Nasal polyps (benign growths on the inside of the nasal lining).
 - Injuries to the nose.
 - Physical or emotional stress.
 - Foreign bodies (usually in children).
- Sometimes, the bleeding may be coming from further back in the nose (posterior bleed). In those cases, bleeding is usually heavy and difficult to control by pinching the nose only.

First aid to help someone with a Nosebleed:

- Make sure the scene is safe.
- Wear PPE.
- Have the person sit and lean forward.
- Pinch the soft part of the nose on both sides with a clean dressing.
- Place constant pressure on the nostrils for a few minutes until the bleeding stop. If the bleeding continues, press harder.
- Phone your local emergency number if
- You can't stop the bleeding in about 15 minutes.
- The bleeding is heavy, such as gushing blood.
- The injured person has trouble breathing.



Press on both sides in nostrils

➤ **Bleeding from mouth:**

Definition:

Bleeding from the blood vessels of the mouth.

Causes of Mouth bleed:

- Infection. - Anticoagulant medications.
- Oral surgery or diseases of the gums.
- Untreated injuries caused by trauma to the mouth, throat, or chest.

First aid to help someone with bleeding from the mouth:

- Make sure the scene is safe.
- Get the first aid kit.
- Wear PPE.
- If bleeding is coming from the tongue, lip, or cheek and you can reach it easily, pressure with gauze or a clean cloth.

➤ **Eye injury:**

Definition:

Damage or trauma inflicted to the eye by external means. The concept includes both surface injuries and intraocular injuries.

Causes of eye injury:

- Blow to the Eye - Cut or Scratch
- Object in the Eye - Chemical Burns, Radiation

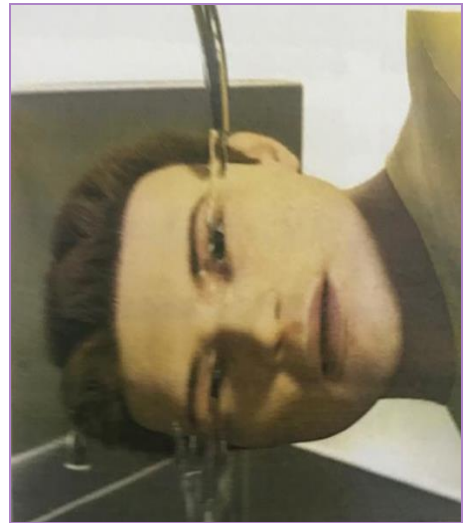
First aid to help someone with an eye injury:

- Make sure the scene is safe.
- Get the first aid kit.
- Wear PPE.

- If someone small like sand in a person`s eye, rinse with lots of running water.
- Phone your local number if:
 - The sand or object doesn`t come out.
 - The person gas extreme pain.
 - The person still has trouble seeing.
 - Tell the person to keep his eye closed until someone with more advanced training arrives and takes over.

First aid to help someone with a toxic eye injury:

- Make sure the scene is safe.
 - Get the first aid kit.
 - Wear PPE.
 - If chemicals get in a person`s eyes, rinse with lots of water. Rinse for at least 15 minutes or until someone with more advanced training arrives and takes over.
- **Caution:** if only one eye is affected, make sure the eye with the chemicals in it the lower eye as you rinse. Try not to rinse the chemicals into the unaffected eye.
 - If an eyewash station is nearby or you have access to an eyewash kit, use it.
 - If neither is available, use water from the tap or normal saline or contact lens solution.
 - Phone your emergency number.



» Internal bleeding:

Internal bleeding is bleeding inside the body.

Causes of internal bleeding:

Trauma: The most common cause of death in trauma is bleeding.

Non-traumatic:

- Blood vessel rupture as a result of high blood pressure, aneurysms, esophageal varicose, peptic ulcers, or ectopic pregnancy.
- Other diseases linked to internal bleeding include cancer, hematologic disease, Vitamin K deficiency, and rare viral hemorrhagic fevers.

Signs of internal bleeding:

- Pain in the abdomen or chest after an injury.
- Shortness of breath after an injury.
- Coughed up or vomited blood after an injury.
- Signs of shock without external bleeding.
- A knife or a gunshot wound.

First aid to help a person with suspected internal bleeding:

- Make sure the scene is safe.
- Phone or send someone to phone your local emergency number and get the first aid kit and the AED.
- Wear PPE.
- Have the person lie down and keep still.
- Check for signs of shock.
- Give CPR if needed.

» Head, neck, and spinal injuries:

Causes of Head, neck, and spinal injuries:

Anything that puts too much pressure or force on the head, neck, or back can result in injury. Common causes are **falls, accidents, and hard blows**.

Signs & Symptoms of Head, neck, and spinal injuries:

- The scalp, neck, or back bleeds.
- It looks like the head, neck, or back is in an odd position.
- Pain is felt in the back, neck, and/or head. The pain can be severe.
- Stiff neck.
- Abdominal pain, Vomiting.
- Blood or fluid comes from the mouth, nose, or an ear.
- Loss of vision, Blurred or double vision, Pupils is not the same size.
- Inability to move any part of the body.
- Weakness in an arm or leg. Walking is difficult.
- New feelings of numbness occur in the legs, arms, shoulders, or any other part of the body.
- New loss of bladder or bowel control occurs.
- Confusion, Drowsiness.
- Convulsions & Loss of consciousness.

First aid to help a person with a possible head, neck, or spinal injury:

- Make sure the scene is safe.
- Phone or send someone to phone your local emergency number.
- Have the person remain as still as possible. Wait for someone with more advanced training to arrive and take over.
- Do not twist or turn the person's head or neck unless absolutely necessary.



» **Burns:**

Definition of Burns:

Tissue damage caused by fires or prolonged exposure to sunlight or any other kind of radiation, as well as contact with hot surfaces or chemicals.

Causes of burns:

- Dry heat
 - Chemical burn
 - Ionizing radiation
- Moist heat
 - Electrical burn
 - Friction

Degree of burns:

- **I degree burns:** minor epithelial damage, present as redness, clinically significant.
- **II-degree burns:** partial thickness burns, scalds. These burns are painful.
- **III-degree burns:** full thickness burns, usually painless.

First aid to take for small burns:

- Make sure the scene is safe.
- Get the first aid kit.
- Wear PPE.
- Cool the burn area immediately with cold, but not ice-cold, water for at least 10 minutes.
- If you do not have cold water, use a coil or cold, but not freezing, clean compress.
- Run cold water on the burn until it doesn't hurt.
- You may cover the burn with a dry, nonstick sterile or clean dressing.



First aid to take for large burns:

- Make sure the scene is safe.
- If there is a fire, the burn area is large, or you're not sure what to do, phone your local emergency number.
- If the person or his clothing is on fire, put the fire out. Have the person stop, drop, and roll. Then, cover the person with a wet blanket.
- Once the fire is out, remove the wet blanket. Carefully remove jewelry and clothing that is not stuck to the skin.
- For large burns, cool the burn area immediately with cold water for at least 10 minutes.
- After you cool the burns, cover them with dry, nonstick, sterile or clean dressings.
- Cover the person with a dry blanket.
- Check for signs of shock.
- A person with a large burn should be seen by a healthcare provider as soon as possible.
- A healthcare provider can determine if additional treatment is necessary.

➤ Environment emergencies

➤ Bites and Stings:

- Bites may cause injuries ranging from superficial scratches to extensive wounds and often become infected with bacteria from the mouth of the biting creature.
- Bites and stings caused by insects such as spiders, scorpions, and fire ants can cause local reactions, but on occasion more serious systemic reactions can occur.

Signs and symptoms that suggest a more serious reaction include:

1. Nausea or vomiting
2. Severe pain at the site
3. Abdominal pain
4. Difficulty breathing
5. Muscle rigidity
6. Headache
7. Decreased responsiveness

First aid to take for an animal or human bite:

- Make sure the scene is safe for you and the person who has been bitten.
- With animal bites, be sure to wash the wound with plenty of soap and water.
- Apply a bag of ice and water wrapped in a towel to help with bruising and swelling.
- If there is a bruise or swelling, place a bag of ice and water wrapped in a towel on the bite for up to 20 minutes.
- For all bites that break the skin, phone a healthcare provider as soon as possible.

➤ **Heat-related emergency:**

➤ **Dehydration:**

Definition:

Dehydration is a loss of too much fluid from the body.

Causes of dehydration:

- Diarrhea - Vomiting
- Excessive sweating - Fever
- Burns
- Diabetes when blood sugar is too high

- Increased urination (undiagnosed diabetes)
- Not drinking enough water, especially on warm and hot days or during or after exercise
- Some medications (diuretics, blood pressure meds, and anticonvulsant medications).

Signs and Symptoms of Dehydration:

- Dry mouth and tongue with thick saliva.
- Flushed face
- Confusion in older adults
- Dry, warm skin - Dizziness
- General weakness - Headache
- Dark, concentrated urine - Decreased urination
- Few or no tears when crying - Decreased skin elasticity

First aid to take for dehydration:

- The best first aid for dehydration is prevention: make sure a person drinks and eats enough to stay hydrated.

➤ Heat cramps:

Heat cramps are painful muscle spasms, most often occurring in the calves, arms, stomach muscles, and back.

Causes of heat cramps:

Intense physical activity in a hot location can lead to heat cramps. The cause is a combination of your rising body temperature and a loss of electrolytes through sweat.

Symptoms of heat cramps:

- Clammy skin.
- Dizziness.
- Tiredness or weakness.
- Heavy sweating.
- Nausea or vomiting.

First aid to help a person who has a heat cramp:

- Get the first aid kit.
- Wear PPE.
- Have the person rest and cool off.
- Have the person drink something with sugar and electrolytes, such as a sports drink or juice, or water if these aren't available.
- If the person can tolerate it, apply a bag with ice and water wrapped in a towel to the cramping area for up to 20 minutes.

➤ Heat stroke:

Definition:

Heat stroke as a core body temperature that rises above 40 °C, accompanied by hot dry skin and central nervous system abnormalities.

Causes of Heat stroke:

Heat stroke results from exposure to a high environmental temperature or from strenuous exercise.

Signs of heat stroke:

- Confusion.
- Feeling faint or fatigue.
- Dizziness.
- Fainting.
- Nausea or vomiting.

- Muscle cramp.
- Nervous system abnormalities, such as delirium, convulsions, or coma.

First aid to help a person who has heat stroke:

- Phone your local emergency.
- Put the person in cool water up to his neck if possible, or spray him with cool water.
- If the person becomes unresponsive and is not breathing normally or only gasping, give CPR.

➤ Cold-related emergency:

➤ Low body temperature (Hypothermia):

Definition:

Hypothermia is an involuntary drop in body temperature below 35C.

Causes of Low body temperature:

Hypothermia is often caused by exposure to cold weather or immersion in cold water.

Signs of low body temperature:

- Skin that's cool to the touch.
- Shivering, which stops when the body temperature is very low.
- Confusion.
- Skin becomes ice-cold and blue.

First aid to help a person who has hypothermia:

- Make sure the scene is safe for you and the person who has hypothermia.
- Get the person out of the cold.

- Remove wet clothing, pat the person dry, and cover with a blanket.
- Get the first aid kit.
- Phone your local emergency number.
- Put dry clothes on the person, Cover the body and head, but not the face, with blankets, towels, or even newspapers.
- Remain with the person until someone with more advanced training arrives and takes over.
- If the person becomes unresponsive and is not breathing normally or only gasping, give CPR.

Students' self-assessment

1- CPR steps are:

- a. Airway, Breathing, Check pulse
- b. Compressions, Airway, Breathing
- c. Airway, Breathing, Compressions
- d. Airway, check pulse, Breathing.

2- The following are classic signs of hyperglycemia except:

- a. Hunger
- b. Hypertension
- c. thirst
- d. frequent urination.

3- When are breaths given?

- a. After EMS arrives.
- b. Before compressions
- c. With compressions
- d. After compressions

4-The first step for someone with nosebleed is:

- a. call emergency
- b. person sits and lean backward.
- c. person sits and lean forward.
- d. measure blood pressure.

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